Pseudo Projection Operators for Exponentially Stable and Safe Neural Stochastic Control*

Jingdong Zhang[a], Shijie Zhou[b], Qunxi Zhu[b,d,e], Wei Lin[b,c,d,e,*] and Xuerong Mao[f]

[a]I-X and Department of Mathematics, Imperial College London, London, SW7 2AZ, UK
[b]Research Institute of Intelligent Complex Systems, Fudan University, 220 Handan Road, Shanghai, 200438, China
[c]School of Mathematical Sciences, SCMS, and SCAM, Fudan University, 220 Handan Road, Shanghai, 200433, China
[d]Shanghai Artificial Intelligence Laboratory, Shanghai, 200232, China
[e]MOE Frontiers Center for Brain Science and State Key Laboratory of Medical Neurobiology, Fudan University, Shanghai, 200032, China
[f]Department of Mathematics and Statistics, University of Strathclyde, Glasgow, G1 1XH, UK

ARTICLE INFO

Keywords:
Nonlinear feedback control
Continuous stochastic dynamics
Neural networks
Safety guarantee
Exponential stability guarantee

ABSTRACT

We, based on the control barrier functions and the stochastic exponential stability theorem, present a general framework for constructing an exponentially stable and safe (ESS) controller for stabilizing stochastic dynamics and constraining the system state in the required safe region simultaneously. Concretely, we introduce pseudo projection operators (PPOs) for safety and exponential stability with closed forms. The PPOs map the candidate controller to the safe controller space and the exponentially stable controller space, respectively. We then introduce a direct ESS PPO based on a Lyapunov-type function that blows up at the boundary of the safe set, which maps a candidate controller to an exponentially stable and safe controller under verifiable conditions. In addition, our framework is extended to systems with high relative degree to improve the generalization of the PPOs. We further extend the PPO framework to stochastic delay-differential equations, establishing sufficient conditions for exponential stability and safety, together with corresponding closed-form PPO constructions. To automatically design the suitable candidate controllers for general nonlinear stochastic dynamics, we propose an artificial neural network based framework for learning the candidate controllers. We provide two unbiased trace estimators to reduce the computational complexity of the framework, improving its performance in high-dimensional tasks. We demonstrate the efficacy of the proposed framework via numerical studies on several representative stochastic dynamics.

1. Introduction

Stabilizing the stochastic dynamics to the target equilibria is one of the major missions in the field of controlling dynamical systems (4; 46; 38). During the last decades algorithms and software for computing the control policies to stabilize linear or polynomial dynamics have reached a high level of maturity (5; 36), enabling its deployment in engineering applications. As for stabilizing more general and nonlinear dynamical systems, linearization technique around the targeted states is often utilized and thus the existing control policies for linear systems are effective in the vicinity of the targeted equilibria (43). Most of the aforementioned techniques perform fairly well in stabilizing the linear or linearized dynamics, however, they lack provable guarantees for stabilizing nonlinear dynamics initiated far away from the equilibria. To circumvent this problem, leveraging the stochastic stabilization theory (30; 31; 3; 18), adaptive control algorithms have been proposed for stabilizing the stochastic dynamics, such as the Quadratic Programming (QP) (41) and model predictive control (MPC) (32). Although these online algorithms succeed in stabilizing the low-dimensional dynamics, they suffer from the at least second order computational complexity (24) to be applied for the high-dimensional tasks.

An alternative to the online algorithms is to find the controller with explicit form with the artificial neural networks (ANNs) (7; 45; 11; 58). In contrast to the online algorithms, the ANNs parameterize the controller as an analytical function with trainable parameters. Once the parameters are fixed, the explicit form of the controller is obtained and the control value for each state is returned in an instant. To find such efficient controller, the ANN updates the parameters via gradient descent methods until the sufficient conditions for stabilizing the target equilibria are satisfied on the training data. Although the ANNs have remarkable abilities in learning the stabilizing controllers, the major issue is that the ANN-based controllers trained on the finite data suffer from the lack of stability guarantee, i.e., satisfying the expected stability conditions on the whole state space. To remedy the issue, (59) requires the loss function in ANNs to satisfy stronger stability conditions and replace the training data as

*S. Z. is supported by the National Natural Science Foundation of China (NSFC) of China under Grant 62403139. Q. Zhu is supported by the National Key R&D Program of China (No. 2025YFA1016503), by the NSFC (Nos. T2541024, 62406072, and 12171350), and by the STCSM (No. 23YF1402500). W. Lin is supported by the NSFC (Nos. 11925103, 12531018), the STCSM (Nos. 2021SHZDZX0103, 22JC1402500, 22JC1401402, 25JS2810400), and the SMEC (Nos. 2023ZKZD04, 2023KEJI05-72). M. X. would like to thank the Royal Society (WM160014, Royal Society Wolfson Research Merit Award) and the Royal Society of Edinburgh (RSE1832) for their financial support.

*Corresponding author.
j.zhang1@imperial.ac.uk (J. Zhang); sjzhou14@fudan.edu.cn (S. Zhou); qxzhu@fudan.edu.cn (Q. Zhu); wlin@fudan.edu.cn (W. Lin); x.mao@strath.ac.uk (X. Mao)
ORCID(s): 0000-0002-2341-0980 (J. Zhang); 0000-0002-5526-4156 (S. Zhou); 0000-0001-7281-5274 (Q. Zhu); 0000-0002-1863-4306 (W. Lin); 0000-0002-6768-9864 (X. Mao)

the discretization points of the state space, and (7) contains a falsifier to add states that violate the stability conditions into the training data. Both techniques improve the performance of the controllers, but they still cannot provide the rigorous guarantee for the controllers since the ANN itself does not have the ability to process infinite data.

Safety, defined as restricting the state of the controlled system within a particular region, is another major concern in industrial applications such as robotic and automotive systems (54). We take safety into account here because the stabilized SDEs does not possess safety, different from the deterministic systems. Recent advances in the control barrier function (CBF) theory pave a direct way to impose safety for the general nonlinear SDEs (9; 21; 1; 50; 51). Several algorithms have been proposed to find the safe controller based on the CBF theory, such as the QP method in (41) and the Sum-of-Square (SOS) method in (33). Although these algorithms try to obtain controllers with safety guarantee, they either have high computational cost or only consider the safety with probability $p < 1$ rather than the probability one required in the safety-critical applications.

Beyond memoryless stochastic dynamics, time delays naturally arise from sensing, communication, and intrinsic memory effects (19). Such delays lead to stochastic delay-differential equations (SDDEs), for which stability and safety analysis is more complicated because the dynamics depend simultaneously on the current and historical states. Existing SDDE stability results are commonly built upon LaSalle-type (28) or Razumikhin-type (55) arguments, while constructing explicit feedback controllers that enforce stability and safety conditions remains challenging. This motivates us to investigate whether the proposed PPO principle can also be extended to delayed stochastic systems.

Contribution. In this paper, we propose an ANN-based framework for efficiently synthesizing exponentially stable and safe (ESS) controllers for stochastic differential equations (SDEs). The framework provides rigorous almost-sure exponential stability and safety guarantees while maintaining linear computational complexity. We first develop closed-form pseudo projection operators (PPOs) that enforce exponential stability and safety separately. We then establish sufficient conditions for ESS control and introduce a direct ESS PPO that maps a candidate controller to an ESS controller under verifiable conditions. Finally, we train an ANN to parameterize candidate controllers and auxiliary functions, and post-process the trained controller using the proposed PPOs to obtain a controller with rigorous ESS guarantees.

Section 2 introduces model formulations and provides some preliminary results. This section is followed by a presentation of the main contributions of this paper, which are summarized as follows.

- In Section 3, we construct PPOs with explicit forms for exponential stability and safety (see Theorems 2, 4). We further establish a direct ESS construction based on a Lyapunov function that diverges at the boundary of the safe set, and show that the corresponding PPO maps a candidate controller to an ESS controller under suitable conditions (see Theorem 6). To improve the application prospects of the PPOs, we extend our results to systems with high relative degree (see Corollary 1).

- In Section 4, we extend the PPO framework to SDDEs. We establish sufficient LaSalle-type conditions for exponential stabilization and safety of controlled SDDEs, which at the same time guarantee the existence of global solution. We then construct explicit PPOs that enforce these conditions.

- In Section 5, we propose an ANN-based framework for learning candidate controllers and auxiliary functions for the underlying stochastic dynamics. In particular, we jointly identify candidate exponentially stable, safe, and ESS controllers based on Theorems 1, 3, and 5. The learned controller is then post-processed by the corresponding PPOs, with the direct ESS PPO in Theorem 6 providing the rigorous exponentially stable and safe controller. To make the proposed framework feasible for high-dimensional tasks, we present two trace estimators that reduce the computational complexity from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.

- In Section 6, we demonstrate the advantages of the proposed framework over the existing methods via numerical studies on several benchmark problems. We further investigate the energy cost of the ANN controller in high relative degree systems and show an application in synchronizing high-dimensional coupled oscillators. Finally we conclude the paper in Section 7.

Notations. The notations used throughout the paper are fairly standard. Denote by $\|\cdot\|$ the Euclidean norm for a vector. Denote by $\|\cdot\|_{\mathrm{F}}$ the Frobenius norm for a matrix. Denote by $\|\cdot\|_{C(\mathbb{R}^d)}$ the maximum norm for the continuous function in $C(\mathbb{R}^d)$. Denote by $|\cdot|$ the modulus length of a scalar. Denote by $a \ll b$ a positive number a is much smaller than b. For a function $V \in C^2(\mathbb{R}^d)$, denote by ∇ and ∇^2 the gradient operator and the associated second order derivative operator. We let $\cdot^+ = \max(\cdot, 0)$, $\mathbb{E}[\cdot]$ denote expectation and $\mathbf{Tr}(\cdot)$ denote the trace. Denote by $\mathrm{Int}(\Omega)$ the interior of the set Ω. Denote by $a \cdot b$ the inner product of the vectors a, b. We denote by $\mathrm{LLC}(\mathbb{R}^d)$ the space of locally Lipschitz continuous functions over $\mathbb{R}^d$. We denote by $\mathbb{R}_+ = \{x \geq 0, x \in \mathbb{R}\}$ the half space of $\mathbb{R}$.

2. Preliminaries

To begin with, we consider the controlled dynamics with time-varying state $\boldsymbol{x}_t \in \mathbb{R}^d$ and control function $\boldsymbol{u} : \mathbb{R}^d \to \mathbb{R}^m$, $m < d$. The state $\boldsymbol{x}_t$ initiated from $\boldsymbol{x}_0$ follows the stochastic differential equation (SDE) in a general form of

$$\mathrm{d}\boldsymbol{x}_t = \left[\boldsymbol{f}\left(\boldsymbol{x}_t\right) + \boldsymbol{g}\left(\boldsymbol{x}_t\right)\boldsymbol{u}(\boldsymbol{x}_t)\right]\mathrm{d}t + \boldsymbol{\sigma}\left(\boldsymbol{x}_t\right)\mathrm{d}W_t, \quad (1)$$

here the drift term $\boldsymbol{f} : \mathbb{R}^d \to \mathbb{R}^d$, the actuated function $\boldsymbol{g} : \mathbb{R}^d \to \mathbb{R}^{d\times m}$, the control function $\boldsymbol{u}$ and the diffusion term $\boldsymbol{\sigma} : \mathbb{R}^d \to \mathbb{R}^{d\times r}$ are Borel-measurable functions, the W_t is a standard r-dimensional (r-D) Wiener process defined on probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$, where $\mathbb{P} : \mathcal{F} \to [0,1]$ is a probability measure, and filtration $\{\mathcal{F}_t\}_{t\geq 0}$ in $\mathcal{F}$ satisfies the regular conditions. The state $\boldsymbol{x}^*$ with $\boldsymbol{f}(\boldsymbol{x}^*) = \mathbf{0}$ and $\boldsymbol{\sigma}(\boldsymbol{x}^*) = \mathbf{0}$ is called the equilibrium. Without loss of generality, we assume origin $\mathbf{0}$ is the equilibrium of the above system. We require $\boldsymbol{u}(\mathbf{0}) = \mathbf{0}$ such that the controller keeps the original equilibrium.

The following locally Lipschitz continuous condition is required throughout the paper.

Assumption 1. *For every integer $n \geq 1$, there is a number $K_n > 0$ such that*

$$\max\left(\|\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{f}(\boldsymbol{y})\|, \|\boldsymbol{g}(\boldsymbol{x}) - \boldsymbol{g}(\boldsymbol{y})\|_{\mathrm{F}}\right) \leq K_n\|\boldsymbol{x} - \boldsymbol{y}\|,$$
$$\max\left(\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{u}(\boldsymbol{y})\|, \|\boldsymbol{\sigma}(\boldsymbol{x}) - \boldsymbol{\sigma}(\boldsymbol{y})\|_{\mathrm{F}}\right) \leq K_n\|\boldsymbol{x} - \boldsymbol{y}\|,$$

for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$ with $\|\boldsymbol{x}\| \vee \|\boldsymbol{y}\| \leq n$.

Definition 1. *(Itô Derivative Operator) For any twice continuously differentiable functions $l : \mathbb{R}^d \to \mathbb{R}$, the Itô differential operator $\mathcal{L}_{\boldsymbol{u}}$ associated with system (1) acts on l as:*

$$\mathcal{L}_{\boldsymbol{u}} l(\boldsymbol{x}) = \nabla l(\boldsymbol{x}) \cdot [\boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}] + \frac{1}{2}\mathbf{Tr}\left[\boldsymbol{\sigma}(\boldsymbol{x})^\top \nabla^2 l(\boldsymbol{x})\boldsymbol{\sigma}(\boldsymbol{x})\right].$$

Definition 2. *We denote by $\alpha_c(x) = c(1+x)$ a special class of increasing functions parameterised by a constant $c > 0$.*

We define the exponential stability and safety as follows.

Definition 3. *(Exponential Stability) The equilibrium of system (1) is said to be exponentially stable a.s., if* $\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}_t\| < 0$ *a.s. for all $\boldsymbol{x}_0 \in \mathbb{R}^d$.*

Definition 4. *(Safety) The system (1) is said to satisfy a safety constraint if for all $0 \leq t < \tau_{\max}$, $\boldsymbol{x}_t \in \mathcal{C}$ a.s. for every solution $\boldsymbol{x}_t$ initiated from $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$, where the maximal existence time of the solution is defined by*

$$\tau_{\max} = \inf\{t \geq 0 : \limsup_{s\uparrow t} \|x_s\| = \infty\}, \tag{2}$$

with the convention $\inf\emptyset = \infty$, $\mathcal{C} \subset \mathbb{R}^d$ is a specified safe region induced by a twice differentiable safe function $h : \mathbb{R}^d \to \mathbb{R}$ as $\mathcal{C} = \{\boldsymbol{x} : h(\boldsymbol{x}) \geq 0\}$, $\mathrm{Int}(\mathcal{C}) = \{\boldsymbol{x} : h(\boldsymbol{x}) > 0\}$ and $\mathbf{0} \in \mathrm{Int}(\mathcal{C})$.

Remark 1. *The safe set in Definition 4 corresponds to what is commonly referred to as an invariance set (60; 35). This notion was originally investigated by (16) in the context of the nonattainability of certain sets. For the sake of terminological consistency, we refer to such sets as safe sets throughout this paper.*

The following theorem proposes the sufficient conditions for exponential stability (refer to (31)).

Theorem 1. *For the controlled system (1), suppose that there exists a function $V \in C^2(\mathbb{R}^d; \mathbb{R}_+)$ with $V(\mathbf{0}) = 0, \varepsilon\|\boldsymbol{x}\|^p \leq V(\boldsymbol{x})$ for constants $p > 0$, $\varepsilon > 0$, such that*

$$\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}) \leq cV(\boldsymbol{x}), \tag{3}$$

for some $c \in \mathbb{R}$ and all $\boldsymbol{x} \in \mathbb{R}^d\backslash\{\mathbf{0}\}$. Then,

$$\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}_t\| \leq \frac{c}{p} \ \ a.s.,$$

for all initial value $\boldsymbol{x}_0 \in \mathbb{R}^d$. In particular, if $c < 0$, the equilibrium of system (1) is exponentially stable.

Even the SDEs are exponentially stable, they may violate the safety constraint, we illustrate this phenomenon in the following example.

Example 1. *Consider a 1-D SDE as:* $\mathrm{d}x = ax\mathrm{d}t + bx\mathrm{d}W_t$, *where $a < 0$, $|a| > \frac{1}{2}b^2$ and $x_0 > 0$, the zero solution is exponentially stable, but for any $M > 0$ and $t > 0$, $\mathbb{P}(x(t) > M) > 0$.*

Proof. Let $V = \frac{1}{2}x^2$, then we have $\mathcal{L}_{\mathbf{0}}V = \left(a + \frac{b^2}{2}\right)x^2 = \left(2a + b^2\right)V$ with $2a + b^2 < 0$. By Theorem 1, the equilibrium of this SDE is exponentially stable. The solution of this SDE is $x_t = x_0\mathrm{e}^{\left(a-\frac{1}{2}b^2\right)t+bW_t}$. Direct calculation leads to $\mathbb{P}(x(t) > M) = \mathbb{P}\left\{W_t > \frac{1}{b}\left[\log\frac{M}{x_0} - \left(a - \frac{b^2}{2}\right)t\right]\right\} > 0$, which completes the proof. ■

2.1. Problem Statement

We assume that the zero solution of the system (1) is unstable, i.e. $\lim_{t\to\infty}\boldsymbol{x}_t \neq \mathbf{0}$ on some set of positive measures. We aim to stabilize the system to $\mathbf{0}$ using ANN-based feedback controller $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x})$ with rigorous stability guarantee and safety guarantee. The designed algorithm is required to have linear computational complexity such that it is applicable for high-dimensional tasks. We denote the controller with both exponential stability guarantee and safety guarantee by ESS controller.

3. Existence and construction of ESS controller

This section presents our construction of ESS controller for system (1) from any candidate controller.

3.1. Pseudo projection operator for exponential stability controller

We begin with the following construction of stability controller for the SDEs.

Theorem 2. *Given function $V \in C^2(\mathbb{R}^d; \mathbb{R}_+)$ satisfying $V(\mathbf{0}) = 0$ and $\varepsilon\|\boldsymbol{x}\|^p \leq V(\boldsymbol{x})$, if for all $\boldsymbol{x} \in \mathbb{R}^d\backslash\{\mathbf{0}\}$, $\boldsymbol{g}^\top\nabla V \neq \mathbf{0}$, for $c < 0$ and any candidate controller $\boldsymbol{u}$,*

define the pseudo projection operator (PPO) for exponential stability as,

$$\pi_{es}(\boldsymbol{u}, V) \triangleq \boldsymbol{u} - \frac{\left(\mathcal{L}_{\boldsymbol{u}} V - cV\right)^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2} + \delta(\boldsymbol{x})}\boldsymbol{g}^{\top}\nabla V, \tag{4}$$

here $\delta(\boldsymbol{x}) \in C^{1}(\mathbb{R}^{d};\mathbb{R}_{+})$ satisfies $\delta(\mathbf{0}) = \delta$, and $\delta(\boldsymbol{x}) \leq \delta\mathbf{1}_{O(\mathbf{0},\mu)}(\boldsymbol{x})$ for a small pre-defined constant $\delta > 0$ and a neighborhood of the equilibrium $O(\mathbf{0}, \mu)$.

If $\mathcal{L}_{\boldsymbol{u}} V \leq cV$ in $O(\mathbf{0}, \mu)$, then, the projected controller $\pi_{es}(\boldsymbol{u}, V)$ is locally Lipschitz continuous and satisfies equation (3), i.e., $\mathcal{L}_{\pi_{es}} V(\boldsymbol{x}) \leq cV(\boldsymbol{x})$ for any $\boldsymbol{x} \in \mathbb{R}^{d}$, guaranteeing the exponential stability of controlled system (1) under controller $\pi_{es}(\boldsymbol{u}, V)$.

Proof. To begin with, we recall that $\boldsymbol{u}$ is locally Lipschitz by our assumption. Therefore, it suffices to verify the local Lipschitz continuity of $\frac{[\mathcal{L}_{\boldsymbol{u}} V - cV]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2}+\delta(\boldsymbol{x})}\boldsymbol{g}^{\top}\nabla V$, which can be factorised into three components: $\boldsymbol{g}^{\top}\nabla V$, $\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}$, and $\frac{1}{\|\boldsymbol{g}^{\top}\nabla V\|^{2}+\delta(\boldsymbol{x})}$.

First, observe that $\boldsymbol{g}$ is locally Lipschitz continuous and that V is twice differentiable, the term $\boldsymbol{g}^{\top}\nabla V$ and its norm $\|\boldsymbol{g}^{\top}\nabla V\|$ are also locally Lipschitz continuous.

Second, assumption 1 guarantees that $\mathcal{L}_{\boldsymbol{u}} V - cV$ is locally Lipschitz continuous. Furthermore, for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{d}$, we have

$$\begin{aligned}&\left\| \left[\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}) - cV(\boldsymbol{x})\right]^{+} - \left[\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{y}) - cV(\boldsymbol{y})\right]^{+} \right\| \\ &\quad \leq \left\| \mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}) - cV(\boldsymbol{x}) - \left[\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{y}) - cV(\boldsymbol{y})\right] \right\|,\end{aligned}$$

which shows that $\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}$ inherits local Lipschitz continuity from $\mathcal{L}_{\boldsymbol{u}} V - cV$.

For the third term, we only need to consider the Lipschitz continuity of $\frac{1}{\|\boldsymbol{g}^{\top}\nabla V\|^{2}+\delta(\boldsymbol{x})}$. For any $\tilde{\boldsymbol{x}} \in \mathbb{R}^{d}$, there exists $\varepsilon > 0$ such that the neighbourhood $O(\tilde{\boldsymbol{x}}, \varepsilon)$ satisfies

$$C_{l} = \sup_{\boldsymbol{x} \in O(\tilde{\boldsymbol{x}}, \varepsilon)} \|\boldsymbol{g}^{\top}(\boldsymbol{x})\nabla V(\boldsymbol{x})\|,$$

and for any $\boldsymbol{x}, \boldsymbol{y} \in O(\tilde{\boldsymbol{x}}, \varepsilon)$,

$$\begin{aligned}&\left\| \frac{1}{\|\boldsymbol{g}(\boldsymbol{x})^{\top}\nabla V(\boldsymbol{x})\|^{2} + \delta(\boldsymbol{x})} - \frac{1}{\|\boldsymbol{g}(\boldsymbol{y})^{\top}\nabla V(\boldsymbol{y})\|^{2} + \delta(\boldsymbol{y})} \right\| \\ &\qquad \leq \frac{(2C_{l}L_{l} + L_{\delta})\|\boldsymbol{x} - \boldsymbol{y}\|}{l^{2}},\end{aligned}$$

where L_{l} (resp. L_{δ}) is the local Lipschitz constant of $\|\boldsymbol{g}^{\top}\nabla V\|$ (resp. $\delta(\boldsymbol{x})$) on $O(\tilde{\boldsymbol{x}}, \varepsilon)$, and

$$l = \inf_{\boldsymbol{x} \in O(\tilde{\boldsymbol{x}}, \varepsilon)} \|\boldsymbol{g}(\boldsymbol{x})^{\top}\nabla V(\boldsymbol{x})\|^{2} + \delta(\boldsymbol{x}).$$

According to the definition of $\delta(\boldsymbol{x})$, we have $l > 0$.

Combining the above results for the three factors, we conclude that $\pi_{\text{es}}(\boldsymbol{u}, V)$ is locally Lipschitz continuous.

By the Itô derivative operator, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\text{es}}(\boldsymbol{u},V)} V =& \nabla V \cdot (\boldsymbol{f} + \boldsymbol{g}\boldsymbol{u}) + \frac{1}{2}\mathbf{Tr}\left(\boldsymbol{\sigma}^{\top}\nabla^{2} V \boldsymbol{\sigma}\right) \\ &- \nabla V \cdot \left\{ \frac{\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2} + \delta(\boldsymbol{x})}\boldsymbol{g}\boldsymbol{g}^{\top}\nabla V \right\} \\ =& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\boldsymbol{g}^{\top}\nabla V\|^{2}\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2} + \delta(\boldsymbol{x})} \\ \leq& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\boldsymbol{g}^{\top}\nabla V\|^{2}\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2} + \delta\mathbf{1}_{O(\mathbf{0},\varepsilon)}(\boldsymbol{x})}\end{aligned}$$

Then over the space $\mathbb{R}^{d}\backslash O(\mathbf{0}, \mu)$, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\text{es}}(\boldsymbol{u},V)} V \leq& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\boldsymbol{g}^{\top}\nabla V\|^{2}\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2}} \\ =& \mathcal{L}_{\boldsymbol{u}} V - \left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+} \leq cV.\end{aligned}$$

Furthermore, since $\mathcal{L}_{\boldsymbol{u}} \leq cV$ for any $\boldsymbol{x} \in O(\mathbf{0}, \mu)$, the following inequality holds inside the small ball $O(\mathbf{0}, \mu)$,

$$\begin{aligned}\mathcal{L}_{\pi_{\text{es}}(\boldsymbol{u},V)} V \leq& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\boldsymbol{g}^{\top}\nabla V\|^{2}\left[\mathcal{L}_{\boldsymbol{u}} V - cV\right]^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2} + \delta} \\ \leq& cV.\end{aligned}$$

Then we have $\mathcal{L}_{\pi_{\text{es}}(\boldsymbol{u},V)} V \leq cV$ over the whole space $\mathbb{R}^{d}$, combining this result with Theorem 1, the controlled system (1) under controller $\pi_{\text{es}}(\boldsymbol{u}, V)$ is exponentially stable, which completes the proof. ■

Remark 2. *The additional local assumption $\mathcal{L}_{\boldsymbol{u}} V \leq cV$ in $O(\mathbf{0}, \mu)$ is used to guarantee the Lipschitz continuity of the projected controller $\pi_{\text{es}}(\boldsymbol{u}, V)$ in the small neighbourhood $O(\mathbf{0}, \mu)$, and it can be removed when the correction term $\frac{(\mathcal{L}_{\boldsymbol{u}} V - cV)^{+}}{\|\boldsymbol{g}^{\top}\nabla V\|^{2}+\delta(\boldsymbol{x})}\boldsymbol{g}^{\top}\nabla V$ is Lipschitz continuous at the origin. In machine-learning-based control, since the training dataset is finitely sampled from a bounded region, the learned controller is often only guaranteed to satisfy $\mathcal{L}_{\boldsymbol{u}} V \leq cV$ locally, rather than globally. This motivates the use of Theorem 2, which is designed precisely for such a setting.*

Remark 3. *We call π_{es} as pseudo projection operator because for the feedback controller $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x})$, π_{es} is not the exact solution of the classic projection problem defined as follows (12).*

$$\begin{aligned}&\pi^{*}(\boldsymbol{u}) \in \arg\min \|\boldsymbol{u} - \tilde{\boldsymbol{u}}\|_{C(\mathbb{R}^{d})}, \\ &\text{s.t.} \quad \mathcal{L}_{\tilde{\boldsymbol{u}}(\boldsymbol{x})} V(\boldsymbol{x}) \leq cV, \ \forall \boldsymbol{x} \in \mathbb{R}^{d}.\end{aligned}$$

Generally, it is difficult to obtain a closed form of the optimal projection operator π^{*}, and this is the major hurdle for applying projection operation in practice. To remedy this issue, QP method is used to solve the following simplified projection for each state as an alternative (8).

$$\begin{aligned}&\tilde{\pi}(\boldsymbol{u})(\boldsymbol{x}) \in \arg\min \|\boldsymbol{u}(\boldsymbol{x}) - \tilde{u}(\boldsymbol{x})\|, \\ &\text{s.t.} \quad \mathcal{L}_{\tilde{\boldsymbol{u}}} V(\boldsymbol{x}) \leq cV(\boldsymbol{x}), \ \forall \boldsymbol{x} \in \mathbb{R}^{d}.\end{aligned}$$

Although this way can assure the expected condition (3) is satisfied for each $\boldsymbol{x}$, it requires the controller to solve an optimization problem in real time for the current input state $\boldsymbol{x}$. Hence, this alternative method is not efficient in the real world scenarios where the state space is high-dimensional or the computational resource is limited.

Remark 4. *The construction of the PPO relies on the condition $\boldsymbol{g}^\top \nabla V \neq \boldsymbol{0}$ in $\mathbb{R}^d \backslash \{\boldsymbol{0}\}$. However, this condition may fail in the systems with high relative degree (9), potentially impeding the application of the proposed projection operator. We propose an approach to construct the pseudo projection operator for high-degree systems in Section 3.4.*

3.2. PPO for safety controller

We study the safety theory and the corresponding construction of safety controller for SDEs in this section.

Theorem 3. *Let Assumption 1 hold and h be a function specified in Definition 4. Let $B(x) = \frac{1}{h(\boldsymbol{x})}$, $\boldsymbol{x} \in \mathrm{Int}(\mathcal{C})$, assume there exists a constant $c_1 \geq 0$ such that*

$$\mathcal{L}_{\boldsymbol{u}} B(\boldsymbol{x}) \leq c_1 \left(1 + B(\boldsymbol{x})\right) := \alpha_{c_1}(B(\boldsymbol{x})), \forall x \in \mathrm{Int}(\mathcal{C}). \quad (5)$$

Then the system (1) satisfies the safety constraint in Definition 4. Furthermore, if $\mathcal{C}$ is bounded, then for all $t \geq 0$, $\boldsymbol{x}_t \in \mathcal{C}$ a.s., for every solution $\boldsymbol{x}_t$ initiated from $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$. Here, B is called the reciprocal control barrier function (RCBF).

Proof. Under Assumption 1, for every initial condition $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$, system (1) admits a unique maximal local strong solution over $[0, \tau_{\max})$, where $\tau_{\max}$ is defined in Eq. (2). Define $\tau_{\partial\mathcal{C}} = \inf\{t \in [0, \tau_{\max}) : h(x_t) = 0\}$. If $\mathcal{C}$ is bounded, then $\tau_{\max} > \tau_{\partial\mathcal{C}}$ *a.s.*, namely, no internal explosion can occur before hitting the boundary.

Let $Y_t = B(\boldsymbol{x}_t) > 0$ and define the exit time

$$\tau_n = \inf\{t \in [0, \tau_{\max}) : Y_t \geq n\},$$
$$\rho_m = \inf\{t \in [0, \tau_{\max}) : \|\boldsymbol{x}_t\| \geq m,\ m \geq \|\boldsymbol{x}_0\|\}.$$

Then we have $\mathbb{P}(\tau_{\partial\mathcal{C}} \leq t) \leq \mathbb{P}(\tau_n \leq t)$ for every $0 < t < \tau_{\max}$. Applying Itô's formula to $Y_{t\wedge\tau_n\wedge\rho_m}$ gives

$$Y_{t\wedge\tau_n\wedge\rho_m} = Y_0 + M_{t\wedge\tau_n\wedge\rho_m} + \int_0^{t\wedge\tau_n\wedge\rho_m} \mathcal{L}_{\boldsymbol{u}} B(\boldsymbol{x}_s)\mathrm{d}s,$$

where $M_t = \int_0^t \nabla B(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\mathrm{d}W_s$ is a local martingale, and by the local Lipschitz continuity of B and $\boldsymbol{\sigma}$ we have $\mathbb{E}M_{t\wedge\tau_n\wedge\rho_m} = 0$. Using (5) and taking expectations yields

$$\begin{aligned}\mathbb{E}\, Y_{t\wedge\tau_n\wedge\rho_m} &\leq Y_0 + \mathbb{E}\int_0^{t\wedge\tau_n\wedge\rho_m} c_1\left(Y_s + 1\right)\mathrm{d}s \\ &= Y_0 + \int_0^t \mathbb{E}\left[c_1 \mathbf{1}_{\{s\leq(\tau_n\wedge\rho_m)\}}\left(Y_s + 1\right)\right]\mathrm{d}s \quad (6) \\ &\leq Y_0 + c_1\int_0^t \mathbb{E}\left(Y_{s\wedge\tau_n\wedge\rho_m} + 1\right)\mathrm{d}s\end{aligned}$$

Set $y_{n,m}(t) = \mathbb{E}Y_{t\wedge\tau_n\wedge\rho_m} + 1$, then we have,

$$y_{n,m}(t) \leq y_{n,m}(0) + c_1\int_0^t y_{n,m}(s)\mathrm{d}s.$$

By the Grönwall inequality (29), we obtain

$$y_{n,m}(t) \leq y_{n,m}(0)\mathrm{e}^{c_1 t} = (Y_0 + 1)\,\mathrm{e}^{c_1 t}, \quad (7)$$

hence $\mathbb{E}\, Y_{t\wedge\tau_n\wedge\rho_m} \leq (Y_0+1)\,\mathrm{e}^{c_1 t} - 1 = C_t$, where C_t depends on t and Y_0 is independent of n. Since $\rho_m \uparrow \tau_{\max}$ a.s., we have $t \wedge \tau_n \wedge \rho_m \uparrow t \wedge \tau_n \wedge \tau_{\max}$ and hence $Y_{t\wedge\tau_n\wedge\rho_m} \to Y_{t\wedge\tau_n\wedge\tau_{\max}}$ a.s.. By Fatou's lemma, it follows that

$$\mathbb{E}Y_{t\wedge\tau_n\wedge\tau_{\max}} \leq \liminf_{m\to\infty} \mathbb{E}Y_{t\wedge\tau_n\wedge\rho_m} \leq C_t.$$

By Markov's inequality,

$$n\,\mathbb{P}(\tau_n \leq (t \wedge \tau_{\max})) \leq \mathbb{E}\, Y_{t\wedge\tau_n\wedge\tau_{\max}} \leq C_t < \infty.$$

Sending $n \to \infty$ gives $\mathbb{P}(\tau_n \leq t \wedge \tau_{\max}) \to 0$ for every $t > 0$, hence $\mathbb{P}(\tau_{\partial\mathcal{C}} \leq t \wedge \tau_{\max}) = 0$ for all t. By a countable union over $t \in \mathbb{Q}_+$, we get $\mathbb{P}(\tau_{\partial\mathcal{C}} < \tau_{\max}) = 0$, and thus x_t never leaves $\mathrm{Int}(\mathcal{C})$ for $t < \tau_{\max}$ almost surely.

When $\mathcal{C}$ is bounded, any trajectory starting in $\mathrm{Int}(\mathcal{C})$ must hit $\partial\mathcal{C}$ before a finite explosion time. Thus, on the event $\{\tau_{\max} < \infty\}$, we have $\tau_{\partial\mathcal{C}} < \tau_{\max}$. Since the preceding argument establishes that $\mathbb{P}(\tau_{\partial\mathcal{C}} < \tau_{\max}) = 0$, it follows that $\mathbb{P}(\tau_{\max} < \infty) = 0$. Consequently, every solution starting in $\mathrm{Int}(\mathcal{C})$ exists globally and satisfies $\mathbb{P}(\boldsymbol{x}_t \in \mathcal{C}$ for all $t \geq 0) = 1$.

■

According to the proof above, condition (5) excludes finite-time termination of any solution initiated from $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$, even when the controller is unbounded on $\mathrm{Int}(\mathcal{C})$. Next, we present the analytical construction of the controller satisfying the condition (5) under given class-$\mathcal{K}$ function.

Theorem 4. *Given a twice differentiable safe function h, let $B(x) = \frac{1}{h(x)}$, $\boldsymbol{x} \in \mathrm{Int}(\mathcal{C})$, assume that $\boldsymbol{g}^\top(\boldsymbol{x})\nabla B(\boldsymbol{x}) \neq \boldsymbol{0}$ for all $\boldsymbol{x} \in \mathrm{Int}(\mathcal{C})\backslash\{\boldsymbol{0}\}$. For any candidate shifted linear function α_{c_1} with constant $c_1 \geq 0$, and controller $\boldsymbol{u}$, define the PPO for safety in $\mathrm{Int}(\mathcal{C})$ as*

$$\pi_{sf}(\boldsymbol{u}, B) = \boldsymbol{u} - \frac{\left[\mathcal{L}_{\boldsymbol{u}} B - \alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top \nabla B\|^2 + \delta(\boldsymbol{x})}\boldsymbol{g}^\top \nabla B, \quad (8)$$

here $\delta(\boldsymbol{x}) \in C^1(\mathrm{Int}(\mathcal{C}); \mathbb{R}_+)$ satisfies $\delta(\boldsymbol{0}) = \delta$, and $\delta(\boldsymbol{x}) \leq \delta\mathbf{1}_{O(\boldsymbol{0},\mu)}(\boldsymbol{x})$ for a small pre-defined constant $\delta > 0$ and a neighborhood of the equilibrium $O(\boldsymbol{0}, \mu) \subset \mathrm{Int}(\mathcal{C})$.

Then, the projected controller $\pi_{sf}(\boldsymbol{u}, B)$ is locally Lipschitz continuous and enforces the safety condition (5). Consequently, the conclusions of Theorem 3 apply to the controlled system (1) under the projected controller $\pi_{sf}(\boldsymbol{u}, B)$.

Proof. To begin with, we recall that $\boldsymbol{u}$ is locally Lipschitz by our assumption. Therefore, it suffices to verify the local

Lipschitz continuity of $\frac{\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta(\boldsymbol{x})}\boldsymbol{g}^\top\nabla B$, which can be factorised into three components: $\boldsymbol{g}^\top\nabla B$, $\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+$, and $\frac{1}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta(\boldsymbol{x})}$.

First, observe that $\boldsymbol{g}$ is locally Lipschitz continuous and that $B=1/h$ is twice differentiable on $\mathrm{Int}(\mathcal{C})$. Consequently, the term $\boldsymbol{g}^\top\nabla B$ and its norm $\|\boldsymbol{g}^\top\nabla B\|$ are also locally Lipschitz continuous.

Second, since B is twice differentiable on $\mathrm{Int}(\mathcal{C})$, Assumption 1 guarantees that $\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)$ is locally Lipschitz continuous. Furthermore, for any $\boldsymbol{x},\boldsymbol{y}\in\mathrm{Int}(\mathcal{C})$, we have

$$\begin{aligned}&\left\|\left[\mathcal{L}_{\boldsymbol{u}}B(\boldsymbol{x})-\alpha_{c_1}(B(\boldsymbol{x}))\right]^+-\left[\mathcal{L}_{\boldsymbol{u}}B(\boldsymbol{y})-\alpha_{c_1}(B(\boldsymbol{y}))\right]^+\right\|\\&\quad\le\left\|\mathcal{L}_{\boldsymbol{u}}B(\boldsymbol{x})-\alpha_{c_1}(B(\boldsymbol{x}))-\left[\mathcal{L}_{\boldsymbol{u}}B(\boldsymbol{y})-\alpha_{c_1}(B(\boldsymbol{y}))\right]\right\|,\end{aligned}$$

which shows that $\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+$ inherits local Lipschitz continuity from $\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)$.

For the third term, we only need to consider the Lipschitz continuity of $\frac{1}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta(\boldsymbol{x})}$. For any $\tilde{\boldsymbol{x}}\in\mathrm{Int}(\mathcal{C})$, there exists $\varepsilon>0$ such that the neighbourhood $O(\tilde{\boldsymbol{x}},\varepsilon)\subset\mathrm{Int}(\mathcal{C})$ satisfies

$$C_l=\sup_{\boldsymbol{x}\in O(\tilde{\boldsymbol{x}},\mu)}\|\boldsymbol{g}^\top(\boldsymbol{x})\nabla B(\boldsymbol{x})\|+\delta(\boldsymbol{x}),$$

and for any $\boldsymbol{x},\boldsymbol{y}\in O(\tilde{\boldsymbol{x}},\varepsilon)$,

$$\begin{aligned}&\left\|\frac{1}{\|\boldsymbol{g}(\boldsymbol{x})^\top\nabla B(\boldsymbol{x})\|^2+\delta(\boldsymbol{x})}-\frac{1}{\|\boldsymbol{g}(\boldsymbol{y})^\top\nabla B(\boldsymbol{y})\|^2+\delta(\boldsymbol{y})}\right\|\\&\qquad\le\frac{(2C_lL_l+L_\delta)\|\boldsymbol{x}-\boldsymbol{y}\|}{l^2},\end{aligned}$$

where L_l (resp. L_δ) is the local Lipschitz constant of $\|\boldsymbol{g}^\top\nabla B\|$ (resp. $\delta(\boldsymbol{x})$) on $O(\tilde{\boldsymbol{x}},\varepsilon)$, and

$$l=\inf_{\boldsymbol{x}\in O(\tilde{\boldsymbol{x}},\mu)}\|\boldsymbol{g}(\boldsymbol{x})^\top\nabla B(\boldsymbol{x})\|^2+\delta(\boldsymbol{x}).$$

According to the definition of $\delta(\boldsymbol{x})$, we have $l>0$. Combining the above results for the three factors, we conclude that $\pi_{\mathrm{sf}}(\boldsymbol{u},B)$ is locally Lipschitz continuous.

By the Itô derivative operator, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\mathrm{sf}}(\boldsymbol{u},B)}B=&\nabla B\cdot(\boldsymbol{f}+\boldsymbol{g}\boldsymbol{u})+\frac{1}{2}\mathbf{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2B\boldsymbol{\sigma}\right)\\&-\nabla B\cdot\left\{\frac{\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta(\boldsymbol{x})}\boldsymbol{g}\boldsymbol{g}^\top\nabla B\right\}\\=&\mathcal{L}_{\boldsymbol{u}}B-\frac{\|\boldsymbol{g}^\top\nabla B\|^2\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta(\boldsymbol{x})}\\\le&\mathcal{L}_{\boldsymbol{u}}B-\frac{\|\boldsymbol{g}^\top\nabla B\|^2\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta\mathbf{1}_{O(\mathbf{0},\mu)}(\boldsymbol{x})}.\end{aligned}$$

Then over the space $\mathrm{Int}(\mathcal{C})\backslash O(\mathbf{0},\mu)$, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\mathrm{sf}}(\boldsymbol{u},B)}B\le&\mathcal{L}_{\boldsymbol{u}}B-\frac{\|\boldsymbol{g}^\top\nabla B\|^2\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2}\\=&\mathcal{L}_{\boldsymbol{u}}B-\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}B\right]^+\le\alpha_{c_1}B.\end{aligned}$$

Inside the small ball $O(\mathbf{0},\mu)$, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\mathrm{sf}}(\boldsymbol{u},B)}B\le&\mathcal{L}_{\boldsymbol{u}}B-\frac{\|\boldsymbol{g}^\top\nabla B\|^2\left[\mathcal{L}_{\boldsymbol{u}}B-\alpha_{c_1}(B)\right]^+}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta\mathbf{1}_{O(\mathbf{0},\mu)}(\boldsymbol{x})}\\\le&\begin{cases}\alpha_{c_1}(B),\ \text{if }\mathcal{L}_{\boldsymbol{u}}B\le\alpha_{c_1}(B),\\\mathbf{1}_{O(\mathbf{0},\varepsilon)(\boldsymbol{x})}\frac{\delta}{\|\boldsymbol{g}^\top\nabla B\|^2+\delta}\mathcal{L}_{\boldsymbol{u}}B+\alpha_{c_1}(B),\ \text{otherwise,}\end{cases}\\\le&\alpha_{c_1}(B)+\mathbf{1}_{O(\mathbf{0},\mu)}(\boldsymbol{x})M,\\\le&\alpha_{c_1}(B)+M\le\alpha_{c_1+M}(B),\end{aligned}$$

here $M=\max_{\boldsymbol{x}\in O(\mathbf{0},\mu)}\mathcal{L}_{\boldsymbol{u}}B<\infty$ since we have proved $\mathcal{L}_{\boldsymbol{u}}B$ is locally Lipschitz continuous. Now we have proved the PPO in Eq. (8) enforces the safety condition (5).

Combining the above results with Theorem 3, the controlled system (1) under controller $\pi_{\mathrm{sf}}(\boldsymbol{u},B)$ satisfies the safety guarantee in Definition 4. And when $\mathcal{C}$ is bounded, the safety guarantee is strengthened to $\boldsymbol{x}_t\in\mathcal{C}$ *a.s.*, $\forall t\ge0$, for every $\boldsymbol{x}_0\in\mathrm{Int}(\mathcal{C})$ which completes the proof. ■

Remark 5. *Since the controller should not change the target equilibrium of the system (1), we require $\pi_{sf}(\boldsymbol{u},B)(\mathbf{0})=\mathbf{0}$. This can be done by requiring $\nabla B(\mathbf{0})=\mathbf{0}$ because the candidate controller should satisfy $\boldsymbol{u}(\mathbf{0})=\mathbf{0}$. Notice that $\nabla B=-\frac{\nabla h(\mathbf{0})}{h^2(\mathbf{0})}$, it is feasible to modify h as $\tilde{h}$ in the vicinity of the equilibrium s.t. $\mathbf{0}\in\{\tilde{h}(\boldsymbol{x})>0\}=\mathrm{Int}(\mathcal{C})$, and $\nabla\tilde{h}(\mathbf{0})=\mathbf{0}$. Then we just need to replace h with $\tilde{h}$.*

For example, Since $h(\mathbf{0})>0$, we may choose constants $0<r_1<r_2$ such that $\overline{B_{r_2}(\mathbf{0})}\subset\mathrm{Int}(\mathcal{C})$, and $h(\boldsymbol{x})\ge\frac{1}{2}h(\mathbf{0})$, $\boldsymbol{x}\in B_{r_2}(\mathbf{0})$, with r_2 sufficiently small such that $r_2\|\nabla h(\mathbf{0})\|<\frac{1}{2}h(\mathbf{0})$. Let $\chi\in C_c^\infty(B_{r_2}(\mathbf{0}))$ be a smooth cut-off function satisfying $0\le\chi(\boldsymbol{x})\le1$, $\chi(\boldsymbol{x})=1$ for all $\boldsymbol{x}\in B_{r_1}(\mathbf{0})$. Define $\widetilde{p}(\boldsymbol{x})=-\chi(\boldsymbol{x})\nabla h(\mathbf{0})^\top\boldsymbol{x}$, and $\widetilde{h}(\boldsymbol{x})=h(\boldsymbol{x})+\widetilde{p}(\boldsymbol{x})$. Since $\chi\in C_c^\infty(B_{r_2}(\mathbf{0}))$, it follows that $\widetilde{p}\in C_c^\infty(B_{r_2}(\mathbf{0}))$. Moreover, $\nabla\widetilde{p}(\boldsymbol{x})=-\nabla\chi(\boldsymbol{x})\nabla h(\mathbf{0})^\top\boldsymbol{x}-\chi(\boldsymbol{x})\nabla h(\mathbf{0})$. Since $\chi(\mathbf{0})=1$, we obtain $\nabla\widetilde{p}(\mathbf{0})=-\nabla h(\mathbf{0})$. Consequently, $\nabla\widetilde{h}(\mathbf{0})=\nabla h(\mathbf{0})+\nabla\widetilde{p}(\mathbf{0})=\mathbf{0}$. It is easy to verify that $\tilde{h}$ keeps the same boundary of safe set as h, then the modified function $\tilde{h}$ satisfies our requirement.

3.3. PPO for ESS controller

The preceding sections provide separate PPOs for stability and safety. We now present the construction of ESS controllers.

Theorem 5. *Let Assumption 1 hold. Suppose that there exists a function $V \in C^2(\mathrm{Int}(\mathcal{C}); \mathbb{R}_+)$ with $V(\mathbf{0}) = 0, \varepsilon\|\boldsymbol{x}\|^p \le V(\boldsymbol{x})$ for constants $p \ge 2$ and $\varepsilon > 0$, such that*

$$\begin{aligned} &\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}) \le cV(\boldsymbol{x}), \\ &\lim_{\boldsymbol{x}\to\partial\mathcal{C}} V(\boldsymbol{x}) = \infty \end{aligned} \tag{9}$$

for some $c < 0$ and all $\boldsymbol{x} \in \mathrm{Int}(\mathcal{C})\backslash\{\mathbf{0}\}$. Then, for all initial value $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$ we have $\boldsymbol{x}_t \in \mathcal{C}$ a.s. for all $t \ge 0$, and $\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}_t\| \le \frac{c}{p}$ a.s.. Here, the controller is denoted as exponentially stable and safe (ESS) controller.

Proof. First, we show safety before the maximal explosion time. Since $V(x) \to \infty$ as $x \to \partial\mathcal{C}$, the function V can be used as a blow-up barrier for the boundary of $\mathcal{C}$. Moreover, condition (9) gives

$$\mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}) \le cV(\boldsymbol{x}) \le c_1(1+V(\boldsymbol{x})) = \alpha_{c_1}(V(\boldsymbol{x})),\ \forall \boldsymbol{x} \in \mathrm{Int}(\mathcal{C}),$$

for any $c_1 > 0$, since $c < 0$ and $V \ge 0$. Therefore, by the same stopping-time argument as in the proof of Theorem 3, with V in place of the reciprocal barrier function B, the controlled system satisfies the safety constraint in Definition 4. That is, for every initial condition $\boldsymbol{x}_0 \in \mathrm{Int}(\mathcal{C})$, we have $\boldsymbol{x}_t \in \mathcal{C},\ 0 \le t < \tau_{\max},\ a.s.$.

To proceed, we show that the local solution is non-explosive, i.e., $\tau_{\max} = \infty$ a.s.. Let $\rho_m = \inf\{t \ge 0 : \|\boldsymbol{x}(t)\| \ge m\}, m > \|\boldsymbol{x}_0\|$. Applying Itô's formula to $V(\boldsymbol{x}(t \wedge \rho_m))$ and using condition (9), we obtain

$$\mathbb{E}V(\boldsymbol{x}(t\wedge\rho_m)) \le V(\boldsymbol{x}_0) + c\mathbb{E}\left[\int_0^{t\wedge\rho_m} V(\boldsymbol{x}(s))\mathrm{d}s\right]$$

Since $c < 0$ and $V(\boldsymbol{x}) \ge 0$, we have

$$\mathbb{E}V(\boldsymbol{x}(t\wedge\rho_m)) \le V(\boldsymbol{x}_0).$$

When $\rho_m \le t$, we have

$$V(\boldsymbol{x}(t\wedge\rho_m)) \ge \varepsilon\|\boldsymbol{x}(\rho_m)\|^p \ge \varepsilon m^p.$$

By Markov's inequality,

$$\varepsilon m^p \mathbb{P}(\rho_m \le t) \le \mathbb{E}\left[V(\boldsymbol{x}(t\wedge\rho_m))\right] \le V(\boldsymbol{x}_0).$$

It follows that $\mathbb{P}(\rho_m \le t) \le \frac{V(\boldsymbol{x}_0)}{\varepsilon m^p}$, and $\lim_{m\to\infty}\mathbb{P}(\rho_m \le t) \to 0$. Since $t \ge 0$ is arbitrary, we conclude that $\tau_{\max} = \infty$, a.s.. Hence the subsequent argument can be carried out on the whole interval $[0, \infty)$.

Next, we consider $\log V(\boldsymbol{x}(t))$. According to Assumption 1 and Lemma 3.2 in (29), for every initial condition $\boldsymbol{x}_0 \ne \mathbf{0}$, we have $\mathbb{P}(\boldsymbol{x}(t) \ne \mathbf{0},\ \forall t \ge 0) = 1$. Therefore, together with $V(\boldsymbol{x}) \ge \varepsilon\|\boldsymbol{x}\|^p$, we have $V(\boldsymbol{x}(t)) > 0$ a.s. for all $t \ge 0$. Applying Itô's formula to $\log V(\boldsymbol{x}(t))$ again,

$$\begin{aligned} \log V(\boldsymbol{x}(t)) &= \log V(\boldsymbol{x}_0) + \int_0^t \frac{\mathcal{L}_{\boldsymbol{u}}V}{V}\mathrm{d}s \\ &\quad + M_t - \frac{1}{2}\int_0^t \frac{\|\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\|^2}{V(\boldsymbol{x}(s))^2}\mathrm{d}s, \\ &\le \log V(\boldsymbol{x}_0) + ct + M_t - \frac{1}{2}\int_0^t \frac{\|\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\|^2}{V(\boldsymbol{x}(s))^2}\mathrm{d}s, \end{aligned}$$

where

$$M_t = \int_0^t \frac{\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))}{V(\boldsymbol{x}(s))}\mathrm{d}W_s.$$

By the exponential martingale inequality,

$$\mathbb{P}\left(\sup_{0\le t\le n}\left[M_t - \frac{\mu}{2}\int_0^t \frac{\|\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\|^2}{V(\boldsymbol{x}(s))^2}\right] > \frac{2}{\mu}\log n\right) \le \frac{1}{n^2},$$

for $0 < \mu < 1$ and $n \in \mathbb{N}^+$. Define set sequence as

$$A_n = \left\{\sup_{0\le t\le n}\left[M_t - \frac{\mu}{2}\int_0^t \frac{\|\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\|^2}{V(\boldsymbol{x}(s))^2}\right] > \frac{2}{\mu}\log n\right\}.$$

Since $\sum_{n=1}^\infty \mathbb{P}(A_n) = \sum_{n=1}^\infty 1/n^2 < \infty$, by Borel-Cantelli lemma we have $\mathbb{P}(\limsup_{n\to\infty} A_n) = 0$. Define $\Omega_0 = (\limsup_{n\to\infty} A_n)^{\mathrm{c}}$, then we have $\mathbb{P}(\Omega_0) = 1$. Therefore, for every $w \in \Omega_0$ there exists $n_0(w)$ such that

$$M_t \le \frac{\mu}{2}\int_0^t \frac{\|\nabla V(\boldsymbol{x}(s))^\top \boldsymbol{\sigma}(\boldsymbol{x}(s))\|^2}{V(\boldsymbol{x}(s))^2} + \frac{2}{\mu}\log n,$$

for $t \le n$ and $n \ge n_0(w)$.

Then we have

$$\log V(\boldsymbol{x}(t)) \le \log V(\boldsymbol{x}_0) + ct + \frac{2}{\mu}\log n, a.s.,$$

which implies

$$\begin{aligned} &\limsup_{t\to\infty}\frac{1}{t}\log V(\boldsymbol{x}_t) \le c,\ \text{a.s.}, \\ &\to \limsup_{t\to\infty}\frac{1}{t}\log\|\boldsymbol{x}_t\| \le \frac{c}{p}\ \text{a.s.}. \end{aligned}$$

Therefore, the proof is complete. ∎

Theorem 6. *Given function $V \in C^2(\mathrm{Int}(\mathcal{C}); \mathbb{R}_+)$ satisfying $V(\mathbf{0}) = 0$, $\varepsilon\|\boldsymbol{x}\|^p \le V(\boldsymbol{x})$ and $\lim_{\boldsymbol{x}\to\partial\mathcal{C}} V(\boldsymbol{x}) \to \infty$, and $\boldsymbol{g}^\top\nabla V \ne \mathbf{0}$ for all $\boldsymbol{x} \in \mathrm{Int}(\mathcal{C})\backslash\{\mathbf{0}\}$. Define the PPO for ESS controller for $c < 0$ and any candidate controller $\boldsymbol{u}$ as,*

$$\pi_{\mathrm{ESS}}(\boldsymbol{u}, V) \triangleq \boldsymbol{u} - \frac{\left(\mathcal{L}_{\boldsymbol{u}}V - cV\right)^+}{\|\boldsymbol{g}^\top\nabla V\|^2 + \delta(\boldsymbol{x})}\boldsymbol{g}^\top\nabla V, \tag{10}$$

here $\delta(\boldsymbol{x}) \in C^1(\mathrm{Int}(\mathcal{C}); \mathbb{R}_+)$ satisfies $\delta(\mathbf{0}) = \delta$, and $\delta(\boldsymbol{x}) \le \delta\mathbf{1}_{O(\mathbf{0},\mu)}(\boldsymbol{x})$ for a small pre-defined constant $\delta > 0$ and a neighborhood of the equilibrium $O(\mathbf{0}, \mu)$.

If $\mathcal{L}_{\boldsymbol{u}}V \le cV$ in $O(\mathbf{0}, \mu)$, then, the projected controller $\pi_{\mathrm{ESS}}(\boldsymbol{u}, V)$ becomes an ESS controller defined in Theorem 5.

The proof of this theorem is similar to the Theorem 2 and Theorem 4 and is omitted here.

Remark 6. *The stability condition in Theorem 5, namely $\mathcal{L}_{\boldsymbol{u}}V \le cV$ for some $c < 0$, is stronger than the safety condition in Theorem 3, which only requires an upper bound*

on the growth of the barrier function through Eq. (5). An alternative approach to constructing an ESS controller is to compose the two projections π_{es} and π_{sf}. A key question is whether the resulting composite projection preserves the original stability and safety guarantees, which requires stronger conditions. Nevertheless, Theorem 6 provides a more direct and transparent construction of the ESS controller. Therefore, the main framework of this paper relies on the direct ESS projection in Theorem 6.

3.4. PPOs for high relative degree systems

The construction of controllers with safety (resp. exponential stability) relies on $\boldsymbol{g}^\top \nabla B \neq \mathbf{0}$ (resp. $\boldsymbol{g}^\top \nabla V \neq \mathbf{0}$), however, these conditions may fail in the systems with high relative degree. Systems with high relative degree may incur the case $\boldsymbol{g}^\top \nabla B = \mathbf{0}$ (resp. $\boldsymbol{g}^\top \nabla V = \mathbf{0}$), rendering the projection operations inapplicable. In this section, we propose an approach to construct PPOs for such high relative degree systems.

We begin with a safe function $h \in C^2(\mathbb{R}^d; \mathbb{R}_+)$ with safe region $C = \{h \geq 0\}$ and $\boldsymbol{g}^\top \nabla h = \mathbf{0}$ for all $\boldsymbol{x} \in C$. Therefore, the RCBF $B = 1/h$ also has $\boldsymbol{g}^\top \nabla B = \mathbf{0}$ and the control input does not appear in $\mathcal{L}_{\boldsymbol{u}} B$. Then we can derive the safe controller for high relative degree systems as follows.

Corollary 1. *Assume there exists a non-negative function $\hat{B}$ and bounded set $\hat{C}$, such that $\lim_{\boldsymbol{x}\to\partial\hat{C}} \hat{B}(\boldsymbol{x}) \to \infty$, $\boldsymbol{g}^\top \nabla \hat{B}(\boldsymbol{x}) \neq \mathbf{0}$, $\forall \boldsymbol{x} \in \hat{C}\backslash\{\mathbf{0}\}$ with $\mathcal{L}_{\boldsymbol{u}} \hat{B} > \alpha_{c_2}(\hat{B})$, and $\hat{C} \subset C$, here $C = \{h \geq 0\}$. Define the PPO for safety as*

$$\hat{\pi}_{\text{sf}}\left(\boldsymbol{u}, \hat{B}\right) = \boldsymbol{u} - \frac{\left[\mathcal{L}_{\boldsymbol{u}} \hat{B} - \alpha_{c_2}\left(\hat{B}\right)\right]^+}{\|\boldsymbol{g}^\top \nabla \hat{B}\|^2 + \delta(\boldsymbol{x})} \boldsymbol{g}^\top \nabla \hat{B}. \tag{11}$$

Here constant $c_2 > 0$ and $\delta(\boldsymbol{x})$ is the same regularization function as introduced in Theorem 4. Then, under controller $\hat{\pi}_{\text{sf}}(\boldsymbol{u}, \hat{B})$ with the shifted linear function α_{c_2}, we have $\boldsymbol{x}_t \in C$ a.s. if $\boldsymbol{x}_0 \in \text{Int}(\hat{C})$.

Proof. Following the same procedure in proof for Theorem 4, definition (11) implies that $\mathcal{L}_{\hat{\pi}_{\text{sf}}(\boldsymbol{u},\hat{B})} \hat{B} \leq \alpha_{c_2}(\hat{B})$, by Theorem 3, we have $\boldsymbol{x}(t) \in \text{Int}(\hat{C}) \subset \text{Int}(C)$ *a.s.* for all $t \geq 0$. ■

The PPO-based stabilizing controller for systems with high relative degree can be derived analogously and is therefore omitted. We further note that, since $V(\mathbf{0}) = \mathbf{0}$ and $V(\boldsymbol{x}) \geq \varepsilon \|\boldsymbol{x}\|^p$, the term $\boldsymbol{g}^\top \nabla V$ is generally not identically zero for all $\boldsymbol{x} \in \mathbb{R}^d$. Consequently, the case $\boldsymbol{g}^\top \nabla B = 0$ is more commonly encountered, as illustrated in Example 3.

A heuristic procedure for constructing the functions $\hat{B}$ in Corollary 1 was proposed in (41), where a sequence of reciprocal functions is defined recursively starting from $h_0 = h$. However, developing an effective and systematic construction for general systems remains an open problem.

3.5. Illustrating examples

To demonstrate the efficacy of the above-established PPOs in dealing with the stochastic dynamical systems, we provide several examples in this section. Specifically, Example 2 demonstrates the ESS controller may not exist under any functions h and V. Example 3 provides a specific implementation of PPOs in systems with high relative degree.

Example 2. *Consider a 2-D SDE as follows,*

$$\begin{aligned} \mathrm{d}x &= (y + u)\,\mathrm{d}t, \\ \mathrm{d}y &= (-2x - y + u)\,\mathrm{d}t + x\mathrm{d}W_t. \end{aligned}$$

Set $V = \frac{1}{2}\left(x^2 + y^2\right)$, $h = x + 1$, $B = 1/h$, the safe region $C = \{(x, y) : x > -1\}$ is unbounded. By calculation,

$$\mathcal{L}_{\boldsymbol{u}} B \leq \alpha_c(B) \iff u \geq -\alpha_c(B)\, h^2 - y$$

$$\mathcal{L}_u V \leq c_2 V \iff (x + y)\, u \leq xy + y^2 - \frac{1}{2}x^2 + \frac{c_2}{2}\left(x^2 + y^2\right)$$

For $x + y = 0$, the above inequality implies that $0 \leq \frac{2c_2 - 1}{2} x^2$ for all $x > -1$, which is impossible. Hence, we have $\{\mathcal{L}_{\boldsymbol{u}} B \leq \alpha_c(B)\} \cap \{\mathcal{L}_{\boldsymbol{u}} V \leq c_2 V\} = \emptyset$ for any $c > 0$ and $c_2 < 0$.

Example 3. *Consider the following 2-D SDE*

$$\begin{aligned} \mathrm{d}x &= -x\mathrm{d}t + \frac{y}{(y^2 + \varepsilon^2)^{1/4}} \mathrm{d}W_t, \\ \mathrm{d}y &= \left(-2x^2 + y + u\right) \mathrm{d}t, \end{aligned} \tag{12}$$

with $h = 1 - x^2$, $C = \{(x, y) : -1 \leq x \leq 1\}$, and $\boldsymbol{g} = (0, 1)^\top$. For $B = 1/h$, we have $\boldsymbol{g}^\top \nabla B = 0$ and

$$\mathcal{L}_{\boldsymbol{u}} B = \frac{\frac{y^2}{\sqrt{y^2 + \varepsilon^2}}\left(3x^2 + 1\right) - 2x^2 h}{h^3}.$$

For any function α, $\mathcal{L}_{\boldsymbol{u}} B \leq \alpha(B)$ implies that

$$\frac{y^2}{\sqrt{y^2 + \varepsilon^2}} \leq \frac{h^3 \alpha\left(\frac{1}{h}\right) - 2x^2 h}{1 + 3x^2},\ y \in \mathbb{R},\ -1 \leq x \leq 1,$$

which is impossible, hence the original system cannot directly satisfy the safety constraint.

Now we define $\hat{h}(x, y) = 1 - x^4 - (1 + 3x^2)\frac{y^2}{\sqrt{y^2 + \varepsilon^2}}$ and introduce the bounded safe set $\hat{C} = \{(x, y) \in \mathbb{R}^2 : H(x, y) \geq 0\}$, together with the reciprocal barrier function $\hat{B}(x, y) = \frac{1}{h_1(x,y)}$ for $(x, y) \in \text{Int}(\hat{C})$. Since $h_1(x, y) \leq 1 - x^4$, we have $\hat{C} \subseteq C$. A direct calculation gives $\mathbf{g}^\top \nabla \hat{B}(x, y) = \frac{(1+3x^2)y(y^2+2\varepsilon^2)}{h_1^2 (y^2+\varepsilon^2)^{3/2}}$. Consequently, $\mathbf{g}^\top \nabla \hat{B}(x, y) = 0$ if and only if $y = 0$ in $\text{Int}(\hat{C})$. Nevertheless, for $y = 0$, we have $\mathcal{L}_{\boldsymbol{u}} \hat{B}(x, 0) = -\frac{4x^4}{(1-x^4)^2} \leq 0$. Therefore, for every $c > 0$, $\mathcal{L}_{\boldsymbol{u}} \hat{B}(x, 0) < c(1 + \hat{B}(x, 0)) = \alpha_c(\hat{B}(x, 0))$. Here we choose an exponentially stabilizing candidate controller $u = ky$ with $k < -1$. By Corollary 1, the corrresponding projected controller

$$\hat{\pi}_{sf}\left(ky, \alpha_c\right) = ky - \frac{\left(\mathcal{L}_{ky} \hat{B} - 1/\hat{h}\right)^+}{\|\boldsymbol{g}^\top \nabla \hat{B}\|^2 + \delta} \boldsymbol{g}^\top \nabla \hat{B}. \tag{13}$$

Then we have $\mathcal{L}_{\hat{\pi}_{sf}}\hat{B} \le \alpha_c(\hat{B})$ *for all* $\boldsymbol{x} \in \mathrm{Int}(\hat{C})$*, which guarantees the safety of the controlled trajectory.*

We compare the performance of this analytical controller (13) *and the ANN-based ESS controller in Section 6.*

4. PPOs for stochastic delay-differential equations

The preceding results focus on stochastic differential equations without time delay. In this section, we show that the proposed PPO framework can be extended to stochastic delay-differential equations (SDDEs) based on a LaSalle-type stability theorem and a newly established control barrier function theory.

To begin with, we consider the control-affine SDDE

$$\begin{aligned}\mathrm{d}\boldsymbol{x}(t) = {} & \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)\mathrm{d}t \\ & + \boldsymbol{g}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)\boldsymbol{u}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)\mathrm{d}t \\ & + \boldsymbol{\sigma}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)\mathrm{d}W_t,\end{aligned} \tag{14}$$

where $\tau > 0$ is the constant time delay, $\boldsymbol{x}(t) = \xi(t)$ for $t \in [-\tau, 0]$ is the initial function, $\boldsymbol{f} : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}_+ \to \mathbb{R}^d$ is the drift term, $\boldsymbol{g} : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}_+ \to \mathbb{R}^{d\times m}$ is the actuation function, $\boldsymbol{u} : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}_+ \to \mathbb{R}^m$ is the feedback controller, and $\sigma : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}_+ \to \mathbb{R}^{d\times r}$ is the diffusion term. We assume that $\boldsymbol{f}(\boldsymbol{0}, \boldsymbol{0}, t) = \boldsymbol{0}$, $\boldsymbol{\sigma}(\boldsymbol{0}, \boldsymbol{0}, t) = \boldsymbol{0}$, and $\boldsymbol{u}(\boldsymbol{0}, \boldsymbol{0}, t) = \boldsymbol{0}$, such that $\boldsymbol{x}(t) \equiv \boldsymbol{0}$ is an equilibrium of Eq. (15). For brevity, we consider the fully actuated control with actuation function $\boldsymbol{g}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t) = \boldsymbol{I}$ as,

$$\begin{aligned}\mathrm{d}\boldsymbol{x}(t) = {} & [\boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t) + \boldsymbol{u}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)]\mathrm{d}t \\ & + \boldsymbol{\sigma}(\boldsymbol{x}(t), \boldsymbol{x}(t-\tau), t)\mathrm{d}W_t,\end{aligned} \tag{15}$$

but the following results can be extended to general case in Eq. (14) by similar approach as that in Section 3.4.

Assumption 2. *Assume that for every integer* $n \ge 1$*, there is a number* $K_n > 0$ *such that*

$$\begin{aligned}&\|\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, t) - \boldsymbol{f}(\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}, t)\| \vee \|\boldsymbol{u}(\boldsymbol{x}, \boldsymbol{y}, t) - \boldsymbol{u}(\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}, t)\| \\ &\vee \|\boldsymbol{\sigma}(\boldsymbol{x}, \boldsymbol{y}, t) - \boldsymbol{\sigma}(\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}, t)\|_{\mathrm{F}} \le K_n(\|\boldsymbol{x} - \bar{\boldsymbol{x}}\| + \|\boldsymbol{y} - \bar{\boldsymbol{y}}\|),\end{aligned}$$

for any $(\boldsymbol{x}, \boldsymbol{y}), (\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}) \in \mathbb{R}^d \times \mathbb{R}^d$ *with* $\|\boldsymbol{x}\| \vee \|\boldsymbol{y}\| \vee \|\bar{\boldsymbol{x}}\| \vee \|\bar{\boldsymbol{y}}\| \le n$.

For a function $V \in C^{2,1}(\mathbb{R}^d \times \mathbb{R}_+; \mathbb{R}_+)$, define the Itô derivative operator associated with Eq. (15) as

$$\begin{aligned}\mathcal{L}_{\boldsymbol{u}}V(\boldsymbol{x}, \boldsymbol{y}, t) = {} & V_t(\boldsymbol{x}, t) + \nabla V(\boldsymbol{x}, t)^{\top}[\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, t) + \boldsymbol{u}(\boldsymbol{x}, \boldsymbol{y}, t)] \\ & + \frac{1}{2}\mathrm{Tr}\left[\boldsymbol{\sigma}^{\top}(\boldsymbol{x}, \boldsymbol{y}, t)\nabla^2 V(\boldsymbol{x}, t)\boldsymbol{\sigma}(\boldsymbol{x}, \boldsymbol{y}, t)\right].\end{aligned}$$

We note that Assumption 2 imposes neither a linear growth condition on the drift and diffusion terms nor the a priori existence of a unique global solution to Eq. (15) for every admissible initial function $\xi \in C_{\mathcal{F}_0}([-\tau, 0]; \mathbb{R}^d)$. Instead, under the stability or safety conditions introduced below, global existence follows as part of the theoretical result.

The following theorem provides unified sufficient conditions for global existence, asymptotic stabilization, and safety of the controlled SDDE.

Theorem 7. *Suppose that Assumption 2 holds. Assume that there exist a function* $V \in C^{2,1}(\mathcal{X} \times [-\tau, \infty); \mathbb{R}_+)$ *and initial data* $\xi(t)$ *for Eq.* (15) *satisfying one of the following three conditions:*

- *(Condition 1)* $\mathcal{X} = \mathbb{R}^d$, $\xi \in C_{\mathcal{F}_0}([-\tau, 0]; \mathbb{R}^d)$, $V(\boldsymbol{x}, t) \ge \varepsilon\|\boldsymbol{x}\|^p$ *for positive constants* ε *and* $p \ge 2$, $\mathcal{L}_{\boldsymbol{u}}V(\boldsymbol{x}, \boldsymbol{y}, t) \le \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau)$*, where* $a > b > 0$*,* $\gamma(t) \ge 0$*,* $\mathrm{e}^{\lambda t}\gamma(t) \in L^1([0, \infty))$ *for constant* $\lambda > 0$ *with* $\lambda - a + b\mathrm{e}^{\lambda\tau} \le 0$.
- *(Condition 2)* $\mathcal{C} \subset \mathbb{R}^d$ *is a compact safe set with nonempty interior,* $\mathcal{X} = \mathrm{Int}(\mathcal{C})$*,* $\xi \in C_{\mathcal{F}_0}([-\tau, 0]; \mathrm{Int}(\mathcal{C}))$*,* $\lim_{\boldsymbol{x}\to\partial\mathcal{C}} \inf_{t\ge 0} V(\boldsymbol{x}, t) = \infty$*,* $\mathcal{L}_{\boldsymbol{u}}V(\boldsymbol{x}, \boldsymbol{y}, t) \le \gamma(t) + K\left(V(\boldsymbol{x}, t) + V(\boldsymbol{y}, t-\tau)\right)$*, for some* $K > 0$*, and* $\gamma(t) \in L^1([0, \infty))$.
- *(Condition 3) Condition 2 holds and, in addition,* $V(\boldsymbol{x}, t) \ge \varepsilon\|\boldsymbol{x}\|^p$ *for positive constants* ε *and* $p \ge 2$*,* $\mathcal{L}_{\boldsymbol{u}}V(\boldsymbol{x}, \boldsymbol{y}, t) \le \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau)$*, where* $a > b > 0$*, and* $\mathrm{e}^{\lambda t}\gamma(t) \in L^1([0, \infty))$ *for constant* $\lambda > 0$ *with* $\lambda - a + b\mathrm{e}^{\lambda\tau} \le 0$.

Then Eq. (15) *admits a unique global solution under each of the above three conditions. Furthermore, under Condition 1,* $\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}(t; \xi)\| \le -\frac{\lambda}{p}$ *a.s.; under Condition 2,* $\boldsymbol{x}(t; \xi) \in \mathcal{C}$ *a.s. for all* $t \ge 0$*; and under Condition 3,* $\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}(t; \xi)\| \le -\frac{\lambda}{p}$*a.s. and* $\boldsymbol{x}(t; \xi) \in \mathcal{C}$ *a.s. for all* $t \ge 0$.

Proof. We establish the three assertions separately.

Proof for Condition 1. By the local Lipschitz property of the coefficients in Eq. (15), there exists a unique maximal local solution. Moreover, Condition 1 yields $\mathcal{L}_{\boldsymbol{u}}V(\boldsymbol{x}, \boldsymbol{y}, t) \le -aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau) + \gamma(t) \le b\left(V(\boldsymbol{x}, t) + V(\boldsymbol{y}, t-\tau)\right) + \gamma(t)$. Together with $\lim_{\|\boldsymbol{x}\|\to\infty} \inf_{t\ge 0} V(\boldsymbol{x}, t) = \infty$, the same stopping-time argument as in the proof of [(28), Theorem A.1] shows that the maximal local solution is nonexplosive. Hence, Eq. (15) admits a unique global solution.

Given the existence of the global solution, we proceed to prove the exponential stability of the controlled system (15). Let $V_t = V(\boldsymbol{x}(t), t)$, we introduce an auxiliary process as $Z_t = \mathrm{e}^{\lambda t}V(\boldsymbol{x}(t), t) + b\mathrm{e}^{\lambda\tau}\int_{t-\tau}^{t} \mathrm{e}^{\lambda s}V(\boldsymbol{x}(s), s)\mathrm{d}s$. By Itô's formula we have

$$\begin{aligned}\mathrm{d}Z_t &= \mathrm{e}^{\lambda t}(\lambda V_t + \mathcal{L}_{\boldsymbol{u}}V + b\mathrm{e}^{\lambda\tau}V_t - bV_{t-\tau})\mathrm{d}t + \mathrm{e}^{\lambda t}\mathrm{d}M_t \\ &\le \mathrm{e}^{\lambda t}\left[(\lambda - a + b\mathrm{e}^{\lambda\tau})V_t + \gamma(t)\right]\mathrm{d}t + \mathrm{e}^{\lambda t}\mathrm{d}M_t \\ &\le \mathrm{e}^{\lambda t}\gamma(t)\mathrm{d}t + \mathrm{e}^{\lambda t}\mathrm{d}M_t,\end{aligned}$$

here $M_t = \int_0^t \nabla V(\boldsymbol{x}(s), s)^{\top}\boldsymbol{\sigma}(\boldsymbol{x}(s), \boldsymbol{x}(s-\tau), s)\mathrm{d}W_s$ is a local martingale. Since $\mathrm{e}^{\lambda t}\gamma(t) \in L^1([0, \infty))$, according to non-negative semimartingale convergence theorem, we have

$\sup_{t\geq 0} Z_t < \infty$ *a.s.*. Note that $\varepsilon\|\boldsymbol{x}\|^p \leq V_t \leq \mathrm{e}^{-\lambda t} Z_t$, we have

$$\limsup_{t\to\infty} \frac{1}{t}\log\|\boldsymbol{x}\| \leq -\frac{\lambda}{p} a.s..$$

Proof for Condition 2. By the local Lipschitz property of the coefficients in Eq. (15), there exists a unique maximal local solution $\boldsymbol{x}(t;\xi)$ on $t \in [-\tau, \tau_{\max})$, where $\tau_{\max}$ denotes the explosion time as defined in Definition 4.

Define the stopping time $\tau_n = \inf\left\{t \in [0, \tau_{\max}) : V_t \geq n\right\}$. Then we have $\lim_{n\to\infty} \tau_n = \tau_{\partial C}$, where $\tau_{\partial C}$ denotes the first time at which the solution reaches the boundary $\partial\mathcal{C}$. Since $V(\boldsymbol{x}, t) \to \infty$ as $\boldsymbol{x} \to \partial\mathcal{C}$ uniformly in $t \geq 0$, the level sets of V are contained in compact subsets of $\mathrm{Int}(\mathcal{C})$. Hence, the stopped stochastic integral associated with $V_{t\wedge\tau_n}$ is a martingale.

Applying Itô's formula gives

$$\mathbb{E}V_{t\wedge\tau_n} = \mathbb{E}V_0 + \mathbb{E}\int_0^{t\wedge\tau_n} \mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}(s), \boldsymbol{x}(s-\tau), s)\mathrm{d}s.$$

By Condition 2, we obtain

$$\begin{aligned}\mathbb{E}V_{t\wedge\tau_n} &\leq \mathbb{E}V_0 + \int_0^{t\wedge\tau_n} \gamma(s)\mathrm{d}s + K\mathbb{E}\int_0^{t\wedge\tau_n} \left(V_s + V_{s-\tau}\right)\mathrm{d}s \\ &\leq C + 2K\int_0^t \left[\sup_{0\leq r\leq s} \mathbb{E}V_{r\wedge\tau_n}\right]\mathrm{d}s,\end{aligned} \tag{16}$$

where $C = \mathbb{E}V_0 + \int_0^\infty \gamma(s)\mathrm{d}s + K\int_{-\tau}^0 \mathbb{E}V(\xi(s), s)\mathrm{d}s = V_0 + \int_0^\infty \gamma(s)\mathrm{d}s + K\int_{-\tau}^0 V(\xi(s), s)\mathrm{d}s < \infty$. Since the right-hand side of Eq. (16) is nondecreasing in t, we further have

$$\sup_{0\leq r\leq t} \mathbb{E}V_{r\wedge\tau_n} \leq C + 2K\int_0^t \left[\sup_{0\leq r\leq s} \mathbb{E}V_{r\wedge\tau_n}\right]\mathrm{d}s.$$

By Grönwall's inequality,

$$\sup_{0\leq r\leq t} \mathbb{E}V_{r\wedge\tau_n} \leq C\mathrm{e}^{2Kt},\ \forall t \geq 0.$$

On the event $\{\tau_n \leq t\}$, $V_{t\wedge\tau_n} = n$. Therefore,

$$n\mathbb{P}(\tau_n \leq t) \leq \mathbb{E}V_{t\wedge\tau_n} \leq C\mathrm{e}^{2Kt},\ \forall t \geq 0.$$

Letting $n \to \infty$ yields $\mathbb{P}(\tau_{\partial C} \leq t) = 0$, $\forall t \geq 0$. By a countable union over $t \in \mathbb{Q}_+$, $\mathbb{P}(\tau_{\partial C} < \infty) = 0$. Since $\mathcal{C}$ is bounded and the solution cannot explode while remaining in $\mathrm{Int}(\mathcal{C})$, it follows that $\mathbb{P}(\tau_{\max} < \infty) = 0$. Consequently, Eq. (15) admits a unique global solution satisfying $\boldsymbol{x}(t;\xi) \in \mathcal{C}$, $\forall t \geq 0$, a.s.

Proof for Condition 3. Condition 2 guarantees that Eq. (15) admits a unique global solution that remains in $\mathcal{C}$ almost surely. And we can use the same analysis as in the proof of Condition 1 to show the exponential convergence of the controlled system.

■

We next construct PPOs for Eq. (15) to directly enforces Conditions 1-3 in Theorem 7.

Theorem 8. *Given a function V satisfying the corresponding conditions in Theorem 7, assume that $\nabla V(\boldsymbol{x}, t) \neq \boldsymbol{0}$ for all $\boldsymbol{x} \neq \boldsymbol{0}$, $t \geq 0$. For any locally Lipschitz candidate controller $\boldsymbol{u}(\boldsymbol{x}, \boldsymbol{y}, t)$, define the PPOs for stabilization and safety as follows:*

- *(Condition 1) Define $\Phi_{\mathrm{es}}(\boldsymbol{u}, V) \triangleq \mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}, \boldsymbol{y}, t) - \gamma(t) + aV(\boldsymbol{x}, t) - bV(\boldsymbol{y}, t-\tau)$ and the corresponding PPO as*

$$\pi_{\mathrm{es}}(\boldsymbol{u}, V)(\boldsymbol{x}, \boldsymbol{y}, t) = \boldsymbol{u} - \frac{\left[\Phi_{\mathrm{es}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V(\boldsymbol{x}, t)\|^2 + \delta(\boldsymbol{x})}\nabla V(\boldsymbol{x}, t).$$

- *(Condition 2) Define $\Phi_{\mathrm{sf}}(\boldsymbol{u}, V) \triangleq \mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}, \boldsymbol{y}, t) - \gamma(t) - K(V(\boldsymbol{x}, t) + V(\boldsymbol{y}, t-\tau))$ and the corresponding PPO as*

$$\pi_{\mathrm{sf}}(\boldsymbol{u}, V)(\boldsymbol{x}, \boldsymbol{y}, t) = \boldsymbol{u} - \frac{\left[\Phi_{\mathrm{sf}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V(\boldsymbol{x}, t)\|^2 + \delta(\boldsymbol{x})}\nabla V(\boldsymbol{x}, t).$$

- *(Condition 3) Define $\Phi_{\mathrm{ESS}}(\boldsymbol{u}, V) \triangleq \mathcal{L}_{\boldsymbol{u}} V(\boldsymbol{x}, \boldsymbol{y}, t) - \gamma(t) + aV(\boldsymbol{x}, t) - bV(\boldsymbol{y}, t-\tau)$ and the corresponding PPO as*

$$\pi_{\mathrm{ESS}}(\boldsymbol{u}, V)(\boldsymbol{x}, \boldsymbol{y}, t) = \boldsymbol{u} - \frac{\left[\Phi_{\mathrm{ESS}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V(\boldsymbol{x}, t)\|^2 + \delta(\boldsymbol{x})}\nabla V(\boldsymbol{x}, t).$$

To avoid singularity of the PPO controllers at the origin, let $\delta(\boldsymbol{x}) \in C^1(\mathcal{X}; \mathbb{R}_+)$ satisfy $\delta(\boldsymbol{0}) = \delta > 0$ and $\delta(\boldsymbol{x}) \leq \delta\boldsymbol{1}_{O(\boldsymbol{0},\mu)}(\boldsymbol{x})$ for a sufficiently small neighborhood $O(\boldsymbol{0}, \mu)$ in which the corresponding conditions are already satisfied, i.e., $\Phi_{\mathrm{es,sf,ESS}}(\boldsymbol{u}, V) \leq 0$ for $\boldsymbol{x} \in O(\boldsymbol{0}, \mu)$. Then, the PPO controllers are locally Lipschitz continuous over $\mathcal{X}$, and the conclusions under Conditions 1–3 in Theorem 7 hold with the corresponding PPOs.

Proof. We first show the proof of PPO under Condition 1. Since $\boldsymbol{f}, \boldsymbol{u}, \boldsymbol{\sigma}$ are locally Lipschitz continuous and $V \in C^{2,1}(\mathbb{R}^d \times [\tau, \infty); \mathbb{R}_+)$, we can prove the local Lipschitz continuity of $\pi_{\mathrm{es}}(\boldsymbol{u}, V)(\boldsymbol{x}, \boldsymbol{y}, t)$ with the same procedure as the proof in Theorem 2, by considering three components ∇V, $\left[\Phi_{\mathrm{es}}(\boldsymbol{u}, V)\right]^+$ and $\frac{1}{\|\nabla V\|^2 + \delta(\boldsymbol{x})}$ separately.

We proceed to verify that $\mathcal{L}_{\pi_{\mathrm{es}}(\boldsymbol{u},V)} V \leq \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau)$. By the Ito derivative operator, we have

$$\begin{aligned}\mathcal{L}_{\pi_{\mathrm{es}}(\boldsymbol{u},V)} V =& \mathcal{L}_{\boldsymbol{u}} V - \nabla V \cdot \frac{\nabla V\left[\Phi_{\mathrm{es}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V\|^2 + \delta(\boldsymbol{x})} \\ \leq& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\nabla V\|^2\left[\Phi_{\mathrm{es}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V\|^2 + \delta\boldsymbol{1}_{O(\boldsymbol{0},\mu)}(\boldsymbol{x})}.\end{aligned}$$

For $\boldsymbol{x} \in \mathbb{R}^d \backslash O(\boldsymbol{0}, \mu)$ we have

$$\begin{aligned}\mathcal{L}_{\pi_{\mathrm{es}}(\boldsymbol{u},V)} V \leq& \mathcal{L}_{\boldsymbol{u}} V - \frac{\|\nabla V\|^2\left[\Phi_{\mathrm{es}}(\boldsymbol{u}, V)\right]^+}{\|\nabla V\|^2} \\ =& \mathcal{L}_{\boldsymbol{u}} V - \left[\mathcal{L}_{\boldsymbol{u}} V - \gamma(t) + aV(\boldsymbol{x}, t) - bV(\boldsymbol{y}, t-\tau)\right]^+ \\ \leq& \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau).\end{aligned}$$

And since $\Phi_{\text{es}}(\boldsymbol{u}, V) \leq 0$ in $O(\mathbf{0}, \mu)$, we have $\mathcal{L}_{\pi_{\text{es}}V(\boldsymbol{u},V)} \leq \mathcal{L}_{\boldsymbol{u}}V \leq \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau)$ for $\boldsymbol{x} \in O(\mathbf{0}, \mu)$.

Then, the Condition 1 in Theorem 7 holds and we obtain the corresponding stability result.

For Conditions 2 and 3, we can proceed in similar way to verify that $\mathcal{L}_{\Phi_{\text{sf}}(\boldsymbol{u},V)}V \leq \gamma(t) + K(V(\boldsymbol{x}, t) + V(\boldsymbol{y}, t-\tau))$ and $\mathcal{L}_{\Phi_{\text{ESS}}(\boldsymbol{u},V)}V \leq \gamma(t) - aV(\boldsymbol{x}, t) + bV(\boldsymbol{y}, t-\tau)$, respectively.

The only difference here is that $\pi_{\text{sf}}(\boldsymbol{u}, V)$ and $\pi_{\text{ESS}}(\boldsymbol{u}, V)$ are locally Lipschitz continuous over $\mathcal{X} = \text{Int}(\mathcal{C})$ instead of $\mathbb{R}^d$ as required by Assumption 2. With such continuity, we can still guarantee the existence of a unique maximal local solution $\boldsymbol{x}(t; \xi)$ on $t \in [-\tau, \tau_{\max})$. And by the same analysis as the *Proof for Condition 2* in Theorem 7, we can obtain that $\mathbb{P}(\tau_{\partial\mathcal{C}} < \infty) = 0$ for $\tau_{\partial\mathcal{C}} = \inf\{t \in [-\tau, \tau_{\max}) : \boldsymbol{x}(t) \in \partial\mathcal{C}\}$, which guarantees the existence of global solution inside the safe region $\mathcal{C}$. With the satisfaction of Conditions 2&3, and the existence of global solution under PPO controllers, the corresponding results of Conditions 2&3 hold, which completes the proof.

■

Remark 7. *Different from Razumikhin-type stability criteria for SDDEs (55), Conditions 1-3 in Theorem 7 are imposed pointwise on the current and delayed states $(\boldsymbol{x}, \boldsymbol{y})$ and do not require an inequality along the whole historical trajectory. For example, a typical Razumikhin-type criterion requires a condition of the form $\mathbb{E}V(\boldsymbol{x}_t(\theta), t+\theta) \leq p\mathbb{E}V(\boldsymbol{x}_t(0), t)$ for all $\theta \in [-\tau, 0]$, under which a suitable upper bound on the infinitesimal generator $\mathcal{L}V(\boldsymbol{x}_t, t)$ is imposed. How to construct an explicit PPO that enforces general Razumikhin-type stability conditions remains an open question.*

5. ANN-based controller synthesis

In this section, we propose the construction of the ESS controller based on ANNs. For clarity, we present the ANN-based controller synthesis below for the SDE setting in Eq. (1). The framework extends analogously to the SDDE setting by using the stability and safety conditions and the corresponding PPOs established in Section 4. The construction process is divided in two steps:

- In Section 5.1, we propose a heuristic ANN-based framework for jointly identifying candidate exponentially stable, safe, and ESS controllers $\boldsymbol{u}$, based on Theorems 1, 3, and 5. Then we apply PPOs to the learned controller to obtain the rigorous ESS controller according to Theorems 2, 4, and 6. Finally, we extend the framework to the systems with high relative degree according to the Corollary 1.
- In section 5.2, we propose two trace estimators to reduce the computational complexity of the proposed ANN framework from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.

5.1. ANN framework

We utilize ANN to parameterize the state feedback controller as $\boldsymbol{u}_{\theta_1}(\boldsymbol{x}) = \mathbf{ANN}(\boldsymbol{\theta}_1, \boldsymbol{x})$, here $\boldsymbol{\theta}_1$ represent the parameters to be trained. To guarantee the existence of unique solutions of controlled system (1), we require the candidate controller $\boldsymbol{u}_{\theta_1}$ to be locally Lipschitz continuous. This requirement is not overly onerous since ANNs are compositions of linear transformations and Lipschitz continuous activation functions (6), such as ReLU and Tanh. In addition, we expect the control value is small based on the consensus that the ideal controllers should be as small as possible (25). For this purpose, we circumscribe the Lipschitz constant of the ANN controller by adopting the spectral normalization trick (56; 34). The other Lipschitz ANN structures such as (53; 27) can also be used here. Furthermore, we require the candidate controller preserves the target equilibrium of the original system, that is $\boldsymbol{u}(\mathbf{0}) = \mathbf{0}$. To fulfill the requirement, we process the output of the ANN as $\boldsymbol{u}_{\theta_1}(\boldsymbol{x}) = \mathbf{ANN}(\boldsymbol{\theta}_1, \boldsymbol{x}) - \mathbf{ANN}(\boldsymbol{\theta}_1, \mathbf{0})$ or $\boldsymbol{u}_{\theta_1}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x})^\top \mathbf{diag}(\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x})\mathbf{ANN}(\boldsymbol{\theta}_1, \boldsymbol{x})$, here $\mathbf{diag}(\boldsymbol{x})$ is a diagonal matrix with its i-th diagonal element as $\boldsymbol{x}_i$.

Next, we construct a twice continuously differentiable function ANN function $V_{\theta_2}(\boldsymbol{x})$ such that $V_{\theta}(\boldsymbol{x}) \geq 0$ and $V_{\theta}(\mathbf{0}) = 0$ as

$$V_{\theta_2}(\boldsymbol{x}) = \mathbf{ANN}\left(\boldsymbol{\theta}_2, \boldsymbol{x}\right) - \mathbf{ANN}\left(\boldsymbol{\theta}_2, \mathbf{0}\right) + \varepsilon\|\boldsymbol{x}\|^p,$$

where $\mathbf{ANN}(\boldsymbol{\theta}_2, \boldsymbol{x})$ is an input convex neural network (2), L^p regularization term $\varepsilon\|\boldsymbol{x}\|^p$ with hyperparameters $\varepsilon \ll 1$ and $p > 0$ is used to guarantee $V_{\theta}(\boldsymbol{x}) \geq \varepsilon\|\boldsymbol{x}\|^p$. As for the twice continuous differentiable property, we adopt the smoothed ReLU activation function ψ to $\mathbf{ANN}(\boldsymbol{\theta}_2, \boldsymbol{x})$ as

$$\psi(x) = \begin{cases} x - \dfrac{d}{2}, x \geq d \\ \dfrac{2dx^3 - x^4}{2d^3}, 0 < x < d, \\ 0, \ x \leq 0. \end{cases}$$

We employ parameterization $\alpha_{\theta_3}(x) = \text{softplus}(\theta_3)(x+1)$ with θ_3 as a trainable scalar parameter for the increasing functions in Theorem 3.

Given the parameterized functions $\boldsymbol{u}_{\theta_1}$, V_{θ_2} and α_{θ_3}, the mission of finding the ANN-based ESS controller $\boldsymbol{u}$ that satisfies the conditions in Theorems 1, 3 can be recast to solve the following optimization problem restricted in the parameter space $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2\, \boldsymbol{\theta}_3\}$,

$$\begin{aligned} &\arg\min_{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3} \|\boldsymbol{u}_{\boldsymbol{\theta}_1}(\boldsymbol{x})^\top \boldsymbol{R}\boldsymbol{u}_{\boldsymbol{\theta}_1}(\boldsymbol{x})\|_{C(\mathbb{R}^d)} \\ &\text{s.t. } \mathcal{L}_{\boldsymbol{u}}V_{\boldsymbol{\theta}_2}(\boldsymbol{x}) - cV_{\boldsymbol{\theta}_2}(\boldsymbol{x}) \leq 0, \ \boldsymbol{x} \in \mathcal{C} \\ &\qquad \mathcal{L}_{\boldsymbol{u}}\left[\frac{1}{h(\boldsymbol{x})}\right] - \alpha_{\boldsymbol{\theta}_3}\left[\frac{1}{h(\boldsymbol{x})}\right] \leq 0, \ \boldsymbol{x} \in \mathcal{C}. \end{aligned} \tag{17}$$

In (17), $\boldsymbol{R} \succ 0$ is a predefined matrix regularizing elements of the controller, $c < 0$ is a predefined constant related to the exponential decay rate of the controlled system. Although c, ε, p can also be set as trainable parameters,

here we fix them for simplicity. To solve the optimization problem (17), the ANNs update the parameters with gradient descent method (14) w.r.t. the loss function in Definition 5.

Remark 8. *Although the final target is to obtain an ESS controller, we include both stability and RCBF-type safety losses during training. Since the RCBF condition in Theorem 3 is weaker than the direct ESS condition in Theorem 5, this yields a more tractable optimization problem. After training, the direct ESS PPO is applied with a boundary-divergent Lyapunov-type function V to enforce the stronger condition in Theorem 6, thereby providing rigorous exponentially stable and safe guarantees. Specifically, we devise the function as $V = V_{\theta_2} + \rho\left(\frac{1}{h(\boldsymbol{x})} - \frac{1}{h(\mathbf{0})}\right)^2$ for some constant $\rho > 0$.*

Definition 5. *Consider the candidate functions $\boldsymbol{u}_{\theta_1}, V_{\theta_2}, \alpha_{\theta_3}$ for the problem (17) with $\boldsymbol{u} = \boldsymbol{u}_{\theta_1}$. Then, the loss function is defined as*

$$\begin{aligned}\mathbb{L}\left(\theta_1,\theta_2,\theta_3\right) =& \mathbb{E}_{\boldsymbol{x}\sim\mu(C)}\Big(\left|\boldsymbol{u}_{\theta_1}(\boldsymbol{x})^\top R\boldsymbol{u}_{\theta_1}(\boldsymbol{x})\right| \\ &+ \lambda_1\left[\mathcal{L}_{\boldsymbol{u}_{\theta_1}} V_{\theta_2}(\boldsymbol{x}) - cV_{\theta_2}(\boldsymbol{x})\right]^+ \\ &+ \lambda_2\left\{\mathcal{L}_{\boldsymbol{u}_{\theta_1}}\left[\frac{1}{h(\boldsymbol{x})}\right] - \alpha_{\theta_3}\left[\frac{1}{h(\boldsymbol{x})}\right]\right\}^+\Big),\end{aligned}$$

where the state variable $\boldsymbol{x}$ obeys the distribution $\mu(C)$ on safe region C, $\lambda_{1,2} > 0$ are the tunable weight factors. In practice, we apply the following empirical loss function:

$$\begin{aligned}&L\left(\theta_1,\theta_2,\theta_3\right) \\ =&\frac{1}{N}\sum_{i=1}^{N}\Big(\left|\boldsymbol{u}_{\theta_1}(\boldsymbol{x}_i)^\top R\boldsymbol{u}_{\theta_1}(\boldsymbol{x}_i)\right| \\ &+ \lambda_1\left[\mathcal{L}_{\boldsymbol{u}_{\theta_1}} V_{\theta_2}(\boldsymbol{x}_i) - cV_{\theta_2}(\boldsymbol{x}_i)\right]^+ \\ &+ \lambda_2\left\{\mathcal{L}_{\boldsymbol{u}_{\theta_1}}\left[\frac{1}{h(\boldsymbol{x}_i)}\right] - \alpha_{\theta_3}\left[\frac{1}{h(\boldsymbol{x}_i)}\right]\right\}^+\Big),\end{aligned} \tag{18}$$

where $\{\boldsymbol{x}_i\}_{i=1}^N$ are sampled from the distribution $\mu = \mu(C)$.

Remark 9. *There are two main challenges of the current ANN framework. Firstly, the ANNs can only handle finite samples in the training stage due to the limit of computational resources. Hence, even the Monte Carlo loss function $L(\theta_1,\theta_2,\theta_3)$ equals to zero on the training data, it generally cannot guarantee the hard constraints in problem (17) are satisfied on the entire C. This is the major drawback of the existing ANN-based control methods (58; 52). Secondly, the complexity of computing Hessian matrix $\nabla^2 V(\boldsymbol{x})$ is $\mathcal{O}(d^2)$ for d-D state $\boldsymbol{x}$, which causes a bottleneck for application in the high-dimensional systems. Below we show how we remedy these issues in this paper.*

One of our contributions is to address the first challenge by applying the PPO to the ANN-based controllers as $\pi_{\text{ESS}}(\boldsymbol{u}_{\theta_1}, V)$, here we devise the function V in Remark 8.

5.2. Reducing computational complexity

In general, computing $\mathcal{L}_{\boldsymbol{u}} V$ costs $\mathcal{O}(d^2)$, since the term $\text{Tr}[\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}]$ requires computing a Hessian matrix of V. The same is true for computing $\mathcal{L}_{\boldsymbol{u}} h$. However, two techniques can help. Firstly, due to the commutativity of the trace operator, we can get the following unbiased estimate:

$$\begin{aligned}\text{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}\right) &= \mathbb{E}\left(\boldsymbol{\xi}^\top\nabla^2 V\boldsymbol{\sigma}\boldsymbol{\sigma}^\top\boldsymbol{\xi}\right) \\ &= \mathbb{E}\left\{\left[\nabla\left(\boldsymbol{\xi}^\top\nabla V\right)\right]^\top\boldsymbol{\sigma}\boldsymbol{\sigma}^\top\boldsymbol{\xi}\right\},\end{aligned}$$

where $\boldsymbol{\xi}$ is a d-D noise vector with zero mean. The Monte Carlo estimator for the above expectation is known as the Hutchinson's trace estimator (20). Second, if $\boldsymbol{\sigma}$ is a vector independent of $\boldsymbol{x}$, computing gradient of the vector-Jacobian product $\boldsymbol{\sigma}^\top\nabla V$ costs $\mathcal{O}(d)$, and the remaining part is simply a vector-Jacobian product as $\text{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}\right) = \boldsymbol{\sigma}^\top\nabla\left(\boldsymbol{\sigma}^\top\nabla V\right)$. These two techniques can be used in the different cases depending on the shape of the diffusion term $\boldsymbol{\sigma}(\boldsymbol{x}) \in \mathbb{R}^{d\times r}$.

5.2.1. Case 1: $r > 1$.

In this case $\boldsymbol{\sigma}$ is a matrix and we use the former technique to accelerate the training stage. Assuming the cost of evaluating $\text{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}\right)$ is on the order of $\mathcal{O}(Nd^2)$ where N is the size of the dataset, then the cost of computing the following Hutchinson's trace estimator via automatic differentiation which only involve vector-Jacobian products is $\mathcal{O}(NMd)$,

$$\text{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}\right) \approx \frac{1}{M}\sum_{i=1}^{M}\left[\nabla\left(\boldsymbol{\xi}_i^\top\nabla V\right)\right]^\top\boldsymbol{\sigma}\boldsymbol{\sigma}^\top\boldsymbol{\xi}_i, \tag{19}$$

where $M \ll d$ is the number of samples. In practice, M can be tuned to trade off variance and computational cost. As reported in (17; 47), $M = 1$ is already a good choice and we adopt it in our experiments.

Remark 10. *Although the PPOs applied to the learned controller in the post-training stage also involves computing the $\mathcal{L}_u V$, this trick cannot be used here because it does not equal the real trace.*

5.2.2. Case 2: $r = 1$.

Since now $\boldsymbol{\sigma}$ is a vector value function, we can use the latter trick by removing the dependency of $\boldsymbol{\sigma}$ on data $\boldsymbol{x}$. Specifically, after getting the $\boldsymbol{\sigma}(\boldsymbol{x})$ on data $\boldsymbol{x}$, we apply the `detach` operator in Pytorch (37) for $\boldsymbol{\sigma}$ to prevent calculating the gradients w.r.t $\boldsymbol{x}$. Then the computing cost of the original trace reduces to $\mathcal{O}(Nd)$ as follows:

$$\text{Tr}\left(\boldsymbol{\sigma}^\top\nabla^2 V\boldsymbol{\sigma}\right) = \boldsymbol{\sigma}^\top\nabla\left\{[\texttt{detach}(\boldsymbol{\sigma})]^\top\nabla V\right\}. \tag{20}$$

Notice that identity (20) is exact, hence it is an ideal choice when $r = 1$. Since the derivative operator $\mathcal{L}_u$ is calculated in both training and post-training control stage and these two cases always cover at least one stage, we can accelerate the whole framework of the proposed ESS controller.

5.3. Motivation of ANN controller

Example 3 shows that we can obtain the ESS controller by directly applying the PPOs to the manually designed candidate controller, which is constructed based on domain experience. In addition, we can also employ non-parametric based methods to construct the candidate controller, such as the kernel machines. Therefore, a natural question is that why we choose trained ANN as the candidate controllers instead of directly using the non-parametric methods or even the random policies. We note that the mechanism of the proposed PPOs is to force the candidate controller to be restricted in the target controller space. Taking the exponentially stable controller space $\mathcal{U}_{es} = \{\boldsymbol{u} : \mathcal{L}_{\boldsymbol{u}} V \leq cV,\ c < 0\}$ as an example, we expect the decay rate of a controller $\boldsymbol{u} \in \mathcal{U}_{es}$,

$$\gamma = \sup_{\boldsymbol{x} \in \mathcal{X} \setminus \{\mathbf{0}\}} \frac{\mathcal{L}_{\boldsymbol{u}} V}{V} \leq c,$$

is as strong as possible, i.e., $0 \ll c - \gamma$. For any candidate controller and V function, the stability condition may not be satisfied at some region, i.e.,

$$\mathcal{L}_{\boldsymbol{u}} V > cV,\ \boldsymbol{x} \in \mathcal{X}_0 \subset \mathcal{X}.$$

In this case, the projection element under the PPO π_{es} is restricted on the boundary of the $\mathcal{U}_{es}$, that is,

$$\mathcal{L}_{\pi_{\text{es}}(\boldsymbol{u},V)} V = cV,\ \boldsymbol{x} \in \mathcal{X}_0,$$

which leads to $\gamma = c$. Nonetheless, if we train the ANN with stability condition based loss, the learned controller is guided to the target controller space and may have stronger decay rate $\gamma < c$, leading to better control performance. And the PPOs are activated only when the trained controller still violates the stability condition at some region out of the training data.

We further numerically investigate the performance of manually designed, non-parametric and ANN based candidate controllers in Section 6 as a support.

6. Experimental results and analysis

In this section, we first validate the efficiency and advantages of the proposed method over the existing methods with several toy examples, then we further apply our method to high-dimensional tasks. The computing device that we use for conducting our experiments includes a single i7-10870 CPU with 16GB memory, and we train all the parameters with Adam optimizer. The experimental configurations are provided in Appendix.

6.1. Case studies on standard control tasks

We evaluate the quality of the proposed ANN-based ESS controller (ESS-PPO) on several case studies. We first show that our approach is able to produce reliable control policies with stability and safety guarantees for each task. We then show the advantages of ESS-PPO in decision time over the other data-efficient algorithms.

6.1.1. Models

We consider two classic control models: the fully actuated double pendulum swing-up and the fully actuated planar kinetic bicycle motion. The double pendulum is a 4-D system with 2 angles and 2 angular velocities: $(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$ (13). We add multiplicative noise force to the system such that the perturbation of angular acceleration is related to angles. The objective is to learn a control strategy that steers the double pendulum to the inverted position within the safe region: the inner pendulum can only spin in $[-\pi/6, 7\pi/6]$, see Figure 1. The kinematic bicycle model for car-like vehicles has 4 variables (x, y, θ, v), where (x, y) is the bicycle's position, θ is the direction of motion and v is the velocity (39). The goal is to steer the noise-perturbed bicycle to the target position without crossing the round wall with a radius of 2m, see Figure 1.

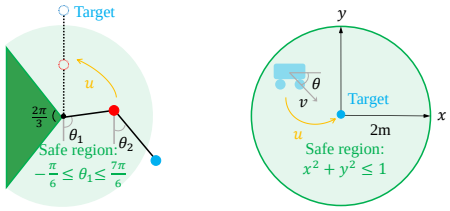


Figure 1: Schematic diagrams of the double pendulum (**left**) and the kinetic bicycle (**right**) with state constraints.

6.1.2. Baselines

To simulate the trajectories of the controlled system, we approximate the solution of the SDE (1) with the Euler-Maruyama method (42). Then the obtained MDPs (or discrete dynamics) can be handled with model-based RL methods. We first compare our ESS-PPO with the online method including GP-MPC algorithm (22) and the BALSA algorithm (15), which are respectively the most data-efficient MPC-based and QP-based RL algorithms to date. Then we compare the ESS-PPO with the existing neural network based offline controllers, including SYNC algorithm (59), NNDM controller (NNDMC) (33) and improved RSM controller (IRSMC) based on the input convex neural network (57).

We conduct 10 independent experiments for each algorithm. In the following figures, the solid lines are obtained through averaging the sampled trajectories, while the shaded areas stand for the variance regions.

6.1.3. Performance.

For the double pendulum task, as shown in Figure 2, all three methods satisfy the safety constraint. Both the ESS-PPO and GP-MPC can steer the pendulum to the inverted position quickly, but the GP-MPC cannot balance the pendulum abidingly whereas the ESS-PPO does. The BALSA has poor performance due to the predefined potential function V and safe function h may induce disjoint safe controllers space and stable controllers space, as illustrated in Example 2. The results in kinetic bicycle task show the same

situations as double pendulum in Figure 3. The motion trajectories of the bicycle under the ESS-PPO are the most robust.

We further compare the performance of the learning based controllers using the kinetic bicycle model. The results provided in Table 1 demonstrate the superiority of the ESS-PPO over the learning based methods in terms of guaranteeing stability and safety, accelerating the training speed and computational efficiency. All the prior methods suffer from the curse of dimensionality, i.e., the k^d term in the computational complexity, for utilizing spatial discretization to numerically obtaining the stability or safety guarantee for neural network based controllers. By contrast, we are the first to endow theoretical stability and safety guarantee to neural controllers with only linear computational complexity. Therefore, our method is applicable to realistic scenarios wherein the dimension of considered system is rather high.

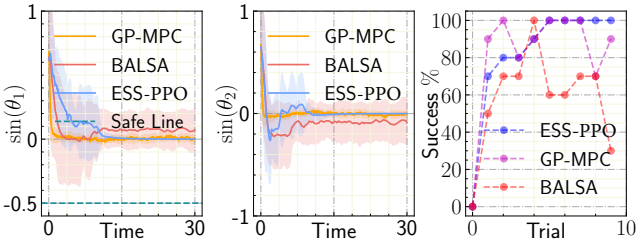


Figure 2: Results of the constraint double pendulum. Sine value along the trajectories of θ_1 (resp., θ_2). Safe region is $\sin(\theta_1) \leq \frac{1}{2}$ (**left** (resp., **center**)). Success rate of three methods, 3 seconds for per trial (**right**). We define 'success' if the pendulum tip's angle is closer than $\frac{\pi}{40}$ to the upright position for consecutive 3 seconds.

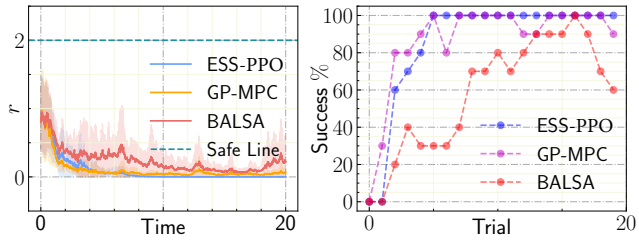


Figure 3: Results of the constraint kinematic bicycle model. The distance r between the car and the target position. Safe region is $r \leq 2$ (**left**). Success rate of three methods, 3 seconds for per trial (**right**). We define 'success' if the car is closer than 0.1 to the target position for consecutive 2 seconds .

6.1.4. Computation Time.

We compare the computation time for one complete control process using all the algorithms. The results are summarized in Table 2. We can see the GP-MPC needs much more time than the others. This is expected since the GP-MPC method has to roll out the dynamics for multiple steps to obtain the predictive states and adjoint states, and then find the optimal control through gradient descent. The BALSA is slightly quicker than ESS-PPO, but its performance is far inferior to that of the latter. Moreover, QP-based solver costs at least $\mathcal{O}(d^2)$ for d-D systems (24), which makes it hard to be applied in the high-dimensional system, whereas our ESS-PPO does. We further underpin this result via an application on a 100-D dynamics (FHN) in the next section.

Table 1
Comparison of learning-based controllers in terms of safety rate, success rate, training time, and the computational complexity wherein d is the system's dimension and k is the size of spatial discretization .

Method	Safety rate (%)	Success rate (%)	Training time	Computational complexity
ESS-PPO	100	100	72s	$\mathcal{O}(d)$
SYNC	90	90	125s	$\mathcal{O}(k^d d^2)$
NNDMC	100	60	209s	$\mathcal{O}(k^d d^3)$
IRSMC	90	90	987s	$\mathcal{O}(k^d)$

Table 2
Average amount of time in seconds needed to complete a control process for each method.

Method	Double Pendulum	Bicycle
ESS-PPO	12.46s	17.34s
GP-MPC	145.81s	56.96s
BALSA	3.17s	4.79s
SYNC	2.28s	1.17s
NNDMC	309.22s	202.65s
IRSMC	3.45s	1.19s

6.2. Application in high-dimensional task

The numerical experiments in the previous section demonstrated the feasibility of the proposed framework. Here, we introduce a practical application of ESS-PPO in synchronizing coupled oscillators. Phase synchronization phenomenon in coupled oscillatory systems have been extensively studied

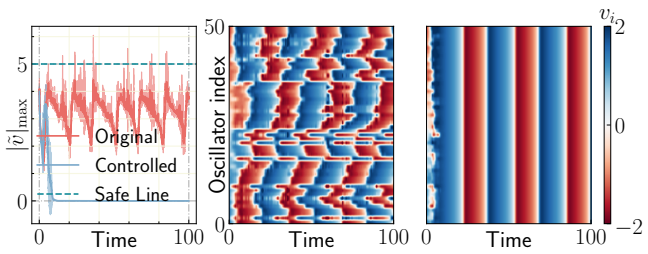


Figure 4: The maximum value of the variances' modulus $|\tilde{v}|_{\max} \triangleq \max_{1\leq i\leq 50} |v_i|$ along the trajectories of the original dynamics and the controlled dynamics (**left**). The solid lines are obtained through averaging the 5 sampled trajectories, while the shaded areas stand for the variance regions. Dynamics of the small-world network of coupled FHN without control (**center**) and with control (**right**). The color represents the membrane potential state v_i of each oscillator

in the context of collective behavior of complex networks from all scales and domains (40; 44; 48). In this section, we show that the ESS-PPO can be used to synchronize the incoherent coupled oscillators in the stochastic settings. Specifically, we consider a small world network of n coupled FitzHugh-Nagumo (FHN) models $\boldsymbol{x} = (v, w)$ (10) under the stochastic coupled forces as:

$$\begin{aligned} \mathrm{d}v_i &= \left(v_i - \frac{v_i^3}{3} - w_i + 1\right)\mathrm{d}t + \frac{1}{3}\sum_{i=1}^{n} L_{ij} v_j \mathrm{d}W_t, \\ \mathrm{d}w_i &= 0.1\left(v_i + 0.7 - 0.8 w_i\right)\mathrm{d}t, \end{aligned}$$

where the fast variable v and the slow variable w correspond to the membrane potential and the recovery variable in an excitable nerve cell, respectively (23). Although the single FHN model has a stable limit cycle Γ, the synchronization manifold $\mathcal{M} \triangleq \left\{\boldsymbol{x}_1(t) \subset \Gamma : \boldsymbol{x}_i(t) = \boldsymbol{x}_1(t), \forall i\right\}$ of the coupled oscillators is not stable due to the stochasticity. We now stabilize the system to the synchronization manifold $\mathcal{M}$ by stabilizing the equilibrium of the dynamics of variance $\tilde{\boldsymbol{x}}_i = \boldsymbol{x}_i - \boldsymbol{x}_1$ with the ESS-PPO. The safe region is $\{\tilde{\boldsymbol{x}}_{1:n} : \max_{1\le i\le n}(\tilde{v}_i, \tilde{w}_i) \le 5\}$ under the safe function $h(\tilde{\boldsymbol{x}}_{i:n}) = 25 - \max_{1\le i\le n}\left(\tilde{v}_i^2, \tilde{w}_i^2\right)$. The results are shown in Figure 4. We can see that the controlled coupled FHN dynamics converge to the synchronization manifold quickly without crossing the safe region. The experimental details are provided in Appendix.

6.3. System with high relative degree

Finally, we consider the case in system with high relative degree modeled by SDE (12). We compare the ANN-based ESS controller with the delicately designed controller (13) in terms of convergence performance and the energy cost in the controlled process. We also consider the non-parametric Kernel based method (57), denoted by Ker, and the improved corresponding version with ESS guarantee by using our PPOs, denoted by ESS+Ker. For convenience, we denote the controller (13) as the linear controller with different strength k (LCk). The energy cost, defined as $\mathcal{E}(t) = \int_0^t \|\boldsymbol{u}(s)\|^2 \mathrm{d}s$, is a crucial factor to measure the quality of a controller (26; 49). Specifically, here we study $\mathcal{E} = \mathcal{E}(5)$. To train the ANN controller, we consider the CBF B_1 with degree 1 constructed in Example 3.

The results are summarized in Figure 5. The energy cost of the ANN controller is lower than that of ESS+Ker and lower than those of most values of k. The energy cost of ANN controller is between those of LC3 and LC5, but the convergence performance (such as variance) of ANN controller is the best. Although the original kernel based controller diverges the dynamics, the improved version ESS+Ker with our PPOs succeed in stabilizing the dynamics without crossing the safe region. Compared to the LC, the contour of ANN controller becomes more dense and rugged as y increases. The learned function V_{θ_2} is significantly different from the conventional Lyapunov function $\frac{1}{2}(x^2 + y^2)$. These results underpin that the ANNs can find more suitable candidate controller than manual design and non-parametric methods.

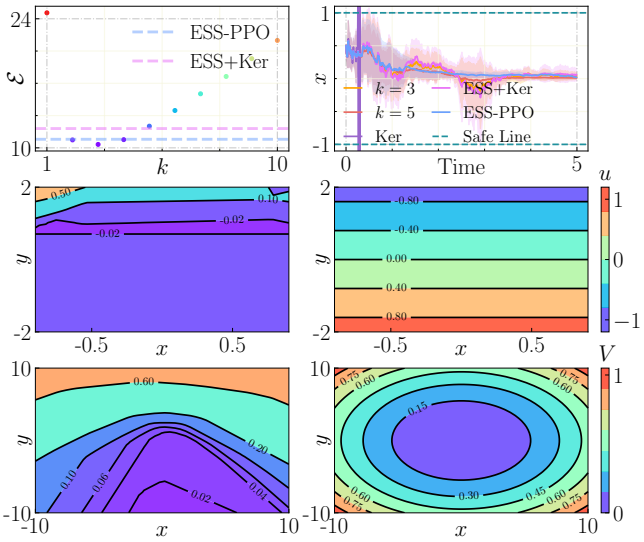


Figure 5: The average energy cost of ANN controller, ESS+Ker controller and the linear controller with $k = 1, \cdots, 10$ over 5 sampled trajectories (**top left**). The time trajectories of the ANN controller and the linear controller with $k = 3, 5$ (**top left**). The solid lines represent the mean value of sampled trajectories, while the shaded areas stand for the variance regions. The contour map of the ANN controller (**middle left**) and the linear controller (**middle right**). The contour map of the learned function V_{θ_2} (**bottom left**) and the function $V = (x^2 + y^2)/2$ (**bottom right**).

7. Conclusion and Future Works

This paper developed an artificial neural network (ANN) based framework to efficiently synthesize exponentially stable and safe (ESS) controllers for stochastic dynamics. The key step is the construction of closed-form pseudo projection operators that can enforce Lyapunov-type stability and barrier-type safety conditions. In particular, we introduced a direct ESS projection based on a Lyapunov-type function that diverges at the boundary of the safe set, thereby avoiding the need to rely on the sequential composition of separate stability and safety projections. The sequential composition result is included as a complementary construction under additional compatibility assumptions. In addition, we extend the PPOs to the high-relative-degree systems to improve the application capabilities of the PPOs in real-world scenarios. We further extended the PPO principle to stochastic delay-differential equations based on LaSalle-type principles. We utilize ANN to learn the suitable candidate controllers, and then we apply the corresponding PPOs to the learned controllers to obtain ESS controllers. To reduce the computational complexity, we exploit two techniques for accelerating the training and post-training stage, which can be adapted to different situations of the underlying systems.

We believe our work is a step towards developing learning-based control algorithms that can be applied in stability-critical and safety-critical real-world scenarios. An exciting direction for future work would be to extend the existing model-based offline framework to the online learning and model-free cases. In addition, the present work focuses on deterministic controllers acting on the drift term for stochastic systems. Extending the PPO framework to genuinely stochastic control policies that act on diffusion term remains an important direction for future work.

References

[1] Ames, A.D., Coogan, S., Egerstedt, M., Notomista, G., Sreenath, K., Tabuada, P., 2019. Control barrier functions: Theory and applications, in: European Control Conference, IEEE. pp. 3420–3431.

[2] Amos, B., Xu, L., Kolter, J.Z., 2017. Input convex neural networks, in: International Conference on Machine Learning, PMLR. pp. 146–155.

[3] Appleby, J.A., Mao, X., Rodkina, A., 2008. Stabilization and destabilization of nonlinear differential equations by noise. IEEE Transactions on Automatic Control 53, 683–691.

[4] Barmish, B., Corless, M., Leitmann, G., 1983. A new class of stabilizing controllers for uncertain dynamical systems. Siam Journal on Control and Optimization 21, 246–255.

[5] Bemporad, A., Morari, M., Dua, V., Pistikopoulos, E.N., 2002. The explicit linear quadratic regulator for constrained systems. Automatica 38, 3–20.

[6] Bishop, C.M., 1994. Neural networks and their applications. Review of Scientific Instruments 65, 1803–1832.

[7] Chang, Y.C., Roohi, N., Gao, S., 2019. Neural Lyapunov control, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 3245–3254.

[8] Chow, Y., Nachum, O., Duenez-Guzman, E., Ghavamzadeh, M., 2018. A Lyapunov-based approach to safe reinforcement learning, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.

[9] Clark, A., 2021. Control barrier functions for stochastic systems. Automatica 130, 109688.

[10] Conley, C., Smoller, J., 1986. Bifurcation and stability of stationary solutions of the fitz-hugh-nagumo equations. Journal of Differential Equations 63, 389–405.

[11] Dawson, C., Qin, Z., Gao, S., Fan, C., 2022. Safe nonlinear control using robust neural Lyapunov-barrier functions, in: Conference on Robot Learning, PMLR. pp. 1724–1735.

[12] Deenen, D.A., Sharif, B., van den Eijnden, S., Nijmeijer, H., Heemels, M., Heertjes, M., 2021. Projection-based integrators for improved motion control: Formalization, well-posedness and stability of hybrid integrator-gain systems. Automatica 133, 109830.

[13] Deisenroth, M.P., Fox, D., Rasmussen, C.E., 2013. Gaussian processes for data-efficient learning in robotics and control. IEEE Transactions on Pattern Analysis and Machine Intelligence 37, 408–423.

[14] Du, S., Lee, J., Li, H., Wang, L., Zhai, X., 2019. Gradient descent finds global minima of deep neural networks, in: International Conference on Machine Learning, PMLR. pp. 1675–1685.

[15] Fan, D.D., Nguyen, J., Thakker, R., Alatur, N., Agha-mohammadi, A.a., Theodorou, E.A., 2020. Bayesian learning-based adaptive control for safety critical systems, in: International Conference on Robotics and Automation, IEEE. pp. 4093–4099.

[16] Friedman, A., 1975. Stochastic Differential Equations and Applications. Springer, Berlin.

[17] Grathwohl, W., Chen, R.T.Q., Bettencourt, J., Duvenaud, D., 2019. Scalable reversible generative models with free-form continuous dynamics, in: International Conference on Learning Representations.

[18] Guo, Y., Lin, W., Chen, G., 2018. Stability of switched systems on randomly switching durations with random interaction matrices. IEEE Transactions on Automatic Control 63, 21–36.

[19] Hale, J.K., Lunel, S.M.V., 2013. Introduction to functional differential equations. Springer Science & Business Media.

[20] Hutchinson, M.F., 1989. A stochastic estimator of the trace of the influence matrix for laplacian smoothing splines. Communications in Statistics-simulation and Computation 18, 1059–1076.

[21] Jankovic, M., 2018. Robust control barrier functions for constrained stabilization of nonlinear systems. Automatica 96, 359–367.

[22] Kamthe, S., Deisenroth, M., 2018. Data-efficient reinforcement learning with probabilistic model predictive control, in: International Conference on Artificial Intelligence and Statistics, PMLR. pp. 1701–1710.

[23] Keener, J., Sneyd, J., 1998. Mathematical physiology, volume 8 of interdisciplinary applied mathematics. Springer-verlag, New York 200, 400.

[24] Kouzoupis, D., Frison, G., Zanelli, A., Diehl, M., 2018. Recent advances in quadratic programming algorithms for nonlinear model predictive control. Vietnam Journal of Mathematics 46, 863–882.

[25] Lewis, F.L., Vrabie, D., Syrmos, V.L., 2012. Optimal Control. John Wiley & Sons.

[26] Li, A., Cornelius, S.P., Liu, Y.Y., Wang, L., Barabási, A.L., 2017. The fundamental advantages of temporal networks. Science 358, 1042–1046.

[27] Liu, H.T.D., Williams, F., Jacobson, A., Fidler, S., Litany, O., 2022. Learning smooth neural functions via lipschitz regularization, in: ACM Siggraph 2022 Conference Proceedings, Association for Computing Machinery. pp. 1–13.

[28] Mao, X., 2002. A note on the lasalle-type theorems for stochastic differential delay equations. Journal of mathematical analysis and applications 268, 125–142.

[29] Mao, X., 2007. Stochastic differential equations and applications. Elsevier.

[30] Mao, X., Marion, G., Renshaw, E., 2002. Environmental brownian noise suppresses explosions in population dynamics. Stochastic Processes and Their Applications 97, 95–110.

[31] Mao, X., Yin, G.G., Yuan, C., 2007. Stabilization and destabilization of hybrid systems of stochastic differential equations. Automatica 43, 264–273.

[32] Mayne, D.Q., 2014. Model predictive control: Recent developments and future promise. Automatica 50, 2967–2986.

[33] Mazouz, R., Muvvala, K., Babu, A.R., Laurenti, L., Lahijanian, M., 2022. Safety guarantees for neural network dynamic systems via stochastic barrier functions, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.

[34] Miyato, T., Kataoka, T., Koyama, M., Yoshida, Y., 2018. Spectral normalization for generative adversarial networks, in: International Conference on Learning Representations.

[35] Nishimura, Y., Hoshino, K., 2024. Control barrier functions for stochastic systems and safety-critical control designs. IEEE Transactions on Automatic Control 69, 8088–8095.

[36] Parrilo, P.A., 2000. Structured Semidefinite Programs and Semialgebraic Geometry Methods in Robustness and Optimization. California Institute of Technology, California.

[37] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al., 2019. Pytorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.

[38] Pyragas, K., Pyragas, V., Kiss, I., Hudson, J., 2002. Stabilizing and tracking unknown steady states of dynamical systems. Physical Review Letters 89, 244103.

[39] Rajamani, R., 2011. Vehicle Dynamics and Control. Springer Science & Business Media, New York.

[40] Rosenblum, M., Pikovsky, A., Kurths, J., Schäfer, C., Tass, P., 2001. Chapter 9 phase SYNChronization: From theory to data analysis, in: Neuro-informatics and Neural Modelling. North-Holland. volume 4 of *Handbook of Biological Physics*, pp. 279–321.

[41] Sarkar, M., Ghose, D., Theodorou, E.A., 2020. High-relative degree stochastic control Lyapunov and barrier functions. Arxiv Preprint Arxiv:2004.03856 .

[42] Särkkä, S., Solin, A., 2019. Applied Stochastic Differential Equations. volume 10. Cambridge University Press, Cambridge.

[43] Sastry, S., Isidori, A., 1989. Adaptive control of linearizable systems. IEEE Transactions on Automatic Control 34, 1123–1131.

[44] Sauseng, P., Klimesch, W., 2008. What does phase information of oscillatory brain activity tell us about cognitive processes? Neuroscience & Biobehavioral Reviews 32, 1001–1013.

[45] Schlaginhaufen, A., Wenk, P., Krause, A., Dorfler, F., 2021. Learning stable deep dynamics models for partially observed or delayed dynamical systems, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 11870–11882.

[46] Socolar, J.E., Sukow, D.W., Gauthier, D.J., 1994. Stabilizing unstable periodic orbits in fast dynamical systems. Physical Review E 50, 3245.

[47] Song, Y., Garg, S., Shi, J., Ermon, S., 2020. Sliced score matching: A scalable approach to density and score estimation, in: Uncertainty in Artificial Intelligence, PMLR. pp. 574–584.

[48] Strogatz, S., 2004. SYNC: The emerging science of spontaneous order. Physics Today 57, 59–60.

[49] Sun, Y.Z., Leng, S.Y., Lai, Y.C., Grebogi, C., Lin, W., 2017. Closed-loop control of complex networks: A trade-off between time and energy. Physical Review Letters 119, 198301.

[50] Taylor, A., Singletary, A., Yue, Y., Ames, A., 2020. Learning for safety-critical control with control barrier functions, in: Learning for Dynamics and Control, PMLR. pp. 708–717.

[51] Taylor, A.J., Ames, A.D., 2020. Adaptive safety with control barrier functions, in: 2020 American Control Conference, IEEE. pp. 1399–1405.

[52] Tsukamoto, H., Chung, S.J., Slotine, J.J.E., 2021. Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview. Annual Reviews in Control 52, 135–169.

[53] Virmaux, A., Scaman, K., 2018. Lipschitz regularity of deep neural networks: Analysis and efficient estimation. Advances in Neural Information Processing Systems 31.

[54] Wang, L., Ames, A.D., Egerstedt, M., 2016. Multi-objective compositions for collision-free connectivity maintenance in teams of mobile robots, in: 2016 IEEE 55th Conference on Decision and Control, IEEE. pp. 2659–2664.

[55] Xu, H., Mao, X., 2022. Advances in discrete-state-feedback stabilization of highly nonlinear hybrid systems by razumikhin technique. IEEE Transactions on Automatic Control 68, 6098–6113.

[56] Yoshida, Y., Miyato, T., 2017. Spectral norm regularization for improving the generalizability of deep learning. Arxiv Preprint Arxiv:1705.10941 .

[57] Zhang, J., Yang, L., Zhu, Q., Lin, W., 2024. Fessnc: Fast exponentially stable and safe neural controller, in: Forty-first International Conference on Machine Learning.

[58] Zhang, J., Zhu, Q., Lin, W., 2022. Neural stochastic control, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 9098–9110.

[59] Zhang, J., Zhu, Q., Yang, W., Lin, W., 2023. SYNC: Safety-aware neural control for stabilizing stochastic delay-differential equations, in: International Conference on Learning Representations.

[60] Zhou, S., Lin, W., Mao, X., Wu, J., 2023. Generalized invariance principles for stochastic dynamical systems and their applications. IEEE Transactions on Automatic Control 69, 85–99.

A. Appendix

We provide the detailed descriptions for the experimental configurations of the control problems in the main text.

A.1. ANN structures

- For constructing the potential function V, we utilize the ICNN as (1):

$$\begin{aligned} \boldsymbol{z}_1 &= \sigma\left(W_0\boldsymbol{x}+\boldsymbol{b}_0\right), \\ \boldsymbol{z}_{i+1} &= \sigma\left(U_i\boldsymbol{z}_i+W_i\boldsymbol{x}+\boldsymbol{b}_i\right),\ i=1,\cdots,k-1, \\ p(\boldsymbol{x}) &\equiv \boldsymbol{z}_k,\ V(\boldsymbol{x})=\sigma\left[p(\boldsymbol{x})-p(\boldsymbol{0})\right]+\varepsilon\|\boldsymbol{x}\|^2, \end{aligned}$$

  where σ is the smoothed **ReLU** function as defined in the main text, $W_i \in \mathbb{R}^{h_i\times d}$, $U_i \in (\mathbb{R}_+\cup\{0\})^{h_i\times h_{i-1}}$, $\boldsymbol{x}\in\mathbb{R}^d$, and, for simplicity, this ICNN function is denoted by ICNN$(h_0,h_1,\cdots,h_{k-1})$;

- The class-$\mathcal{K}$ function α is constructed as:

$$\begin{aligned} \boldsymbol{q}_{i+1} &= \mathrm{ReLU}\left(W_i\boldsymbol{q}_i+\boldsymbol{b}_i\right), i=0,\cdots,k-2, \\ \boldsymbol{q}_k &= \mathrm{ELU}\left(W_{k-1}\boldsymbol{q}_{k-1}+b_{k-1}\right),\ \alpha(x)=\int_0^x q_k(s)\,\mathrm{d}s, \end{aligned}$$

  where $W_i \in \mathbb{R}^{h_{i+1}\times h_i}$, and this class-$\mathcal{K}$ function is denoted by $\mathcal{K}(h_0,h_1,\cdots,h_k)$;

- The neural control function (nonlinear version) is constructed as:

$$\begin{aligned} \boldsymbol{z}_{i+1} &= \mathcal{F}\left(W_i\boldsymbol{z}_i+\boldsymbol{b}_i\right),\ i=0,\cdots,k-1, \\ \mathbf{NN}(\boldsymbol{x}) &\equiv W_k\boldsymbol{z}_k,\ \boldsymbol{u}(\boldsymbol{x})=\mathrm{diag}(\boldsymbol{x})\,\mathbf{NN}(\boldsymbol{x}), \end{aligned}$$

  where $\mathcal{F}(\cdot)$ is the activation function, $W_i \in \mathbb{R}^{h_{i+1}\times h_i}$, this controller is denoted by Control$(h_0,h_1,\cdots,h_{k+1})$.

A.2. Double Pendulum

Here we model the state of the fully actuated noise-perturbed double pendulum as $\dot{\theta}_{1,2}=z_{1,2}+g_{1,2}u_{1,2}$,
$\mathrm{d}z_1=(m_2g\sin\theta_2\cos(\theta_1-\theta_2)-m_2\sin(\theta_1-\theta_2)[l_1z_1^2\cos(\theta_1-\theta_2)+l_2z_2^2]-(m_1+m_2)g\sin\theta_1)/(l_1m_1+m_2\sin^2(\theta_1\theta_2)+g_3u_3)\mathrm{d}t+\sin(\theta_1)\mathrm{d}W_1(t)$,
$\mathrm{d}z_2=[((m_1+m_2)[l_1z_1^2\sin(\theta_1-\theta_2)-g\sin\theta_2+g\sin\theta_1\cos(\theta_1-\theta_2)]+m_2l_2z_2^2\sin(\theta_1-\theta_2)\cos(\theta_1-\theta_2))/(l_2[m_1+m_2\sin^2(\theta_1-\theta_2)])+g_4u_4]\mathrm{d}t+\sin(\theta_2)\mathrm{d}W_2(t)$.

We transform the equilibrium to the zero by the coordinate transformation $\tilde{\theta}_{1,2}=\theta_{1,2}-\pi$. The safe region $h(\theta_{1,2},z_{1,2})=\sin(\theta_1)+0.5 \iff h(\tilde{\theta}_{1,2},z_{1,2})=0.5-\sin(\tilde{\theta}_1)$. Then we consider the dynamic for $\boldsymbol{x}=(\tilde{\theta}_{1,2},z_{1,2})$, and the control signals $\boldsymbol{u}(\boldsymbol{x})=(u_1(\boldsymbol{x}),u_2(\boldsymbol{x}),u_3(\boldsymbol{x}),u_4(\boldsymbol{x}))$ directly act on $\boldsymbol{x}$ with $\boldsymbol{g}=\mathrm{diag}(g_1,g_2,g_3,g_4)=\mathrm{diag}(0,0,1,1)$. For training controller $\boldsymbol{u}$, we sample 500 data from the safe region $[-7\pi/6,\pi/6]\times[-5,5]^3$, we construct the ANNs as follows.

ESS-PPO We parameterize $V(\boldsymbol{x})$ as ICNN(4, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1,10,10,1)$, $\boldsymbol{u}(\boldsymbol{x})$ as Control(4, 12, 12, 4). We set ε = 1e-3, $c=-0.1$. We train the parameters with lr = 0.1 for 300 steps.

GP-MPC Since we have known the dynamics, there is no need to fitting the dynamics as GP flows from data. We obtain the dynamics of moment $\boldsymbol{z}=(\mu,\Sigma)$ of distribution $p(\boldsymbol{x})$ as

$$\begin{aligned} \mu_{t+1} &= \mu_t+dt\left(\cdot f\left(\mu_t\right)+\boldsymbol{u}_t\right), \\ \Sigma_{t+1} &= \Sigma_t+dt\cdot g\left(\mu_t\right), \end{aligned}$$

where f, g are the drift and diffusion terms, respectively. We denote the above dynamic as

$$\boldsymbol{z}_{t+1}=\tilde{f}\left(\boldsymbol{z}_t,\boldsymbol{u}_t\right).$$

Then we solve the MPC problem as

$$\begin{aligned} \min_{\boldsymbol{u}_{0:k-1}}\ &\Phi\left(\boldsymbol{z}_k\right)+\sum_{i=0}^{k-1}l\left(\boldsymbol{z}_i,u_i\right), \\ &\text{s.t. } \boldsymbol{z}_{i+1}=\tilde{f}\left(\boldsymbol{z}_i,\boldsymbol{u}_i\right),\ \boldsymbol{z}_0=(\boldsymbol{x},\boldsymbol{0}), \end{aligned}$$

and apply the first control element $\boldsymbol{u}_0$ to the current dynamic. We use the PMP method as proposed in (2) to solve the above problem. For safety and stability, we select cost function as

$$\begin{aligned} \Phi(\boldsymbol{z}) &= p_1\|\mu\|^2+p_3\|\Sigma\|^2, \\ l(\boldsymbol{z},\boldsymbol{u}) &= p_2\mathrm{ReLU}\left[\sin\left(\tilde{\theta}_1\right)-0.5\right]+\|\boldsymbol{z}\|^2. \end{aligned}$$

We set $p_1=6, p_2=2.5, p_3=0.5, k=10$ and find the optimal control sequence via the gradient descent of the Hamiltonian function in PMP.

QP The object function in QP is set as

$$\begin{aligned} \min_{\boldsymbol{u},d_1,d_2}\ &\frac{1}{2}\|\boldsymbol{u}\|^2+p_1d_1^2+p_2d_2^2, \\ &\text{s.t.}\mathcal{L}_{\boldsymbol{u}}V-\varepsilon V\le d_1, \\ &\mathcal{L}_{\boldsymbol{u}}\left[\frac{1}{h(\boldsymbol{x})}\right]-\gamma\frac{1}{h(\boldsymbol{x})}\le d_2, \end{aligned}$$

where $d_{1,2}$ are the relaxation numbers. We choose $V=\frac{1}{2}\|\boldsymbol{x}\|^2, p_1=10, p_2=10, \varepsilon=0.5, \gamma=2$.

We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{1,4,6,8,9,14,15,16,17,18\}$.

A.3. Planar kinematic bicycle model

Here we model the state of the common noise-perturbed kinematic bicycle system as $\boldsymbol{x}=(x,y,\theta,v)^\top$, where x, y are the coordinate positions in phase plane, θ is the heading, v is the velocity. The dynamic is as follows,

$$\begin{aligned} \mathrm{d}x(t) &= \left[v(t)\cos\theta(t)+u_1\right]\mathrm{d}t+x(t)\,\mathrm{d}W_1(t) \\ \mathrm{d}y(t) &= \left[v(t)\sin\theta(t)+u_2\right]\mathrm{d}t+y(t)\,\mathrm{d}W_2(t) \\ \mathrm{d}\theta(t) &= \left[v(t)+u_3\right]\mathrm{d}t \\ \mathrm{d}v(t) &= \left[x(t)^2+y(t)^2+u_4\right]\mathrm{d}t \end{aligned}$$

The safe function is $h(x,y,\theta,v)=2^2-(x^2+y^2)$. For training controller $\boldsymbol{u}$, we sample 500 data $(r\cos(w), r\sin(w), \theta, v)$

from the safe region $(r, w, \theta, v) \in [0,3] \times [0, 2\pi] \times [-3,3]^2$, we construct the NNs as follows.

ESS-PPO We parameterize $V(\boldsymbol{x})$ as ICNN(4, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{x})$ as Control(4, 12, 12, 4). We set $\varepsilon =$ 1e-3, $c = -0.5$. We train the parameters with lr $= 0.05$ for 500 steps.

GP-MPC Since we have known the dynamics, there is no need to fitting the dynamics as GP flows from data. We obtain the dynamics of moment $\boldsymbol{z} = (\mu, \Sigma)$ of distribution $p(\boldsymbol{x})$ as $\mu_{t+1} = \mu_t + dt(\cdot f(\mu_t) + \boldsymbol{u}_t)$, $\Sigma_{t+1} = \Sigma_t + dt \cdot g(\mu_t)$, where f, g are the drift and diffusion terms, respectively. We denote the above dynamic as $\boldsymbol{z}_{t+1} = \tilde{f}(\boldsymbol{z}_t, \boldsymbol{u}_t)$. Then we solve the MPC problem as

$$\min_{\boldsymbol{u}_{0:k-1}} \Phi\left(\boldsymbol{z}_k\right) + \sum_{i=0}^{k-1} l\left(\boldsymbol{z}_i, u_i\right),$$
$$\text{s.t. } \boldsymbol{z}_{i+1} = \tilde{f}\left(\boldsymbol{z}_i, \boldsymbol{u}_i\right),\ \boldsymbol{z}_0 = (\boldsymbol{x}, \boldsymbol{0}),$$

and apply the first control element $\boldsymbol{u}_0$ to the current dynamic. We use the PMP method as proposed in (2) to solve the above problem. For safety and stability, we select cost function as

$$\Phi(\boldsymbol{z}) = p_1\|\mu\|^2 + p_3\|\Sigma\|^2,$$
$$l(\boldsymbol{z}, \boldsymbol{u}) = p_2 \text{ReLU}\left(\boldsymbol{z}_x^2 + \boldsymbol{z}_y^2 - 2^2\right) + \|\boldsymbol{z}\|^2.$$

We set $p_1 = 5, p_2 = 2.5, p_3 = 0.5, k = 5$ and find the optimal control sequence via the gradient descent of the Hamiltonian function in PMP.

QP The object function in QP is set as

$$\min_{\boldsymbol{u}, d_1, d_2} \frac{1}{2}\|\boldsymbol{u}\|^2 + p_1 d_1^2 + p_2 d_2^2,$$
$$\text{s.t.} \mathcal{L}_{\boldsymbol{u}} V - \varepsilon V \le d_1, \mathcal{L}_{\boldsymbol{u}}\left[\frac{1}{h(\boldsymbol{x})}\right] - \gamma\frac{1}{h(\boldsymbol{x})} \le d_2,$$

where $d_{1,2}$ are the relaxation numbers. We choose $V = \frac{1}{2}\|\boldsymbol{x}\|^2, p_1 = 10, p_2 = 10, \varepsilon = 0.5, \gamma = 2$.

We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{0, 5, 7, 8, 10, 12, 14, 17, 18, 21\}$.

A.4. Fitzhugh-Nagumo model

Consider the coupled FHN model $\boldsymbol{x}_{1:50}$ as $\mathrm{d}\boldsymbol{x}_i = (f(\boldsymbol{x}_i) + u_i)\mathrm{d}t + \sum_{j=1}^N g(\boldsymbol{x}_j)\mathrm{d}W_t$ where $\boldsymbol{x}_i = v_i, w_i$, $f(\boldsymbol{x}_i) = (v_i - \frac{v_i^3}{3} - w_i + 1, 0.1(v_i + 0.7 - 0.8w_i))^\top$, $g(\boldsymbol{x}_i) = (\frac{1}{3}v_i, 0)^\top$. Let $\boldsymbol{\delta} = (\tilde{\boldsymbol{x}}_1, \cdots, \tilde{\boldsymbol{x}}_{50})^\top = (\boldsymbol{x}_1 - \boldsymbol{s}, \cdots, \boldsymbol{x}_{50} - \boldsymbol{s})^\top$, where $\boldsymbol{s}$ is on the synchronization manifold s.t. $\dot{\boldsymbol{s}} = f(\boldsymbol{s})$, then we have the variance equation $\mathrm{d}\boldsymbol{\delta} = (I \otimes \nabla f)\boldsymbol{\delta}\mathrm{d}t + (I \otimes \nabla g)\boldsymbol{\delta}\mathrm{d}W_t$. We only need to stabilize the variance $\boldsymbol{\delta}$ to zero, and the safe function is set as $h(\boldsymbol{\delta}) = 5^2 - \max_{1\le i\le 50}(\tilde{v}_i^2, \tilde{w}_i^2)$. We parameterize $V(\boldsymbol{\delta})$ as ICNN(100, 100, 100, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{\delta})$ as Control(100, 200, 200, 100). In practice, we select mean field $1/N\sum_{j=1}^N \boldsymbol{x}_j$ as $\boldsymbol{s}$ and obtain controller $\boldsymbol{u}(\boldsymbol{\delta})$, which directly acts on $\boldsymbol{x}$ with $\boldsymbol{g} = \boldsymbol{I}_{100\times 100}$. We set $\varepsilon =$ 1e-3, $c = -0.1$. We train the parameters with lr $= 0.1$ for 300 steps. We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{1, 4, 5, 9, 15\}$.

A.5. Model in Example 3

We adopt the CBF $\hat{B}$ in Example 3 with relative degree 1, and we sample 5000 data from the safe region $\mathcal{C}_2$ as defined in Example 3. We parameterize $V(\boldsymbol{x})$ as ICNN(2, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{x})$ as
Control(2, 6, 6, 2). We set $\varepsilon =$ 1e-3, $c = -2$. We train the parameters with lr $= 0.05$ for 500 steps. We use Euler-Maruyama numerical scheme to simulate the system with ANN controller and LCk with $k = 1, \cdots, 10$, and the random seeds are set as $\{3, 5, 6, 7, 8\}$.

References

[1] B. Amos, L. Xu, and J. Z. Kolter, "Input convex neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, PMLR, pp. 146–155, 2017.

[2] S. Kamthe and M. P. Deisenroth, "Data-efficient reinforcement learning with probabilistic model predictive control," in *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics*, PMLR, pp. 1701–1710, 2018.